1) Lesson 9/10/26

1.1) Inequalities

General Procedure:

1. Determine the domain D.
2. Bring all terms to one side: $f(x) \gtrless 0$.
3. Factor into products or fractions: $\frac{A(x)}{B(x)} \gtrless 0$.
    1. Find the roots of each factor.
    2. Construct a sign chart.
4. Intersect the resulting intervals with the domain D.

There can happen 3 main cases: fractions, square roots or absolute values

Ex c) Fraction

Solve the following inequality:

$$\frac{x-1}{x-2} > \frac{2x-3}{x-3}$$

Domain:

$$D = x \in \mathbb{R} \setminus \{2, 3\}$$

Simplify:

$$\frac{x-1}{x-2} - \frac{2x-3}{x-3} > 0 \implies \frac{(x-1)(x-3) - (2x-3)(x-2)}{(x-2)(x-3)} > 0$$

This is of form $A(x)/B(x)$ so the study of the numerator and denominator needs to be done independently

Numerator:
Find zeroes:

$$\begin{aligned}(x-1)(x-3) - (2x-3)(x-2) &= (x^2 - 4x + 3) - (2x^2 - 7x + 6) \\ &= -x^2 + 3x - 3\end{aligned}$$

Recall the quadratic formula:
Let $a, b, c \in \mathbb{R}$, then $ax^2 + bx + c = 0$ has the following solution:

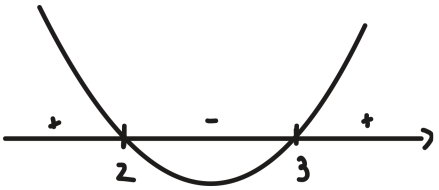
$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where the discriminant $\Delta = b^2 - 4ac$ determines the types of solution:

- $\Delta > 0$: two real (distinct) solutions
- $\Delta = 0$: two real (identical) solutions
- $\Delta < 0$: no real solution

In this case since $\Delta = 9 - 12 = -3 < 0$ we don't have any solutions, hence since $a < 0$ the whole numerator is negative.

Denominator:
Here the function already shows the zeroes, so we can plot the graph to see the regions:



Curve

So we have positive denominator for $x < 2 \cup x > 3$.

By noticing the interval where they have the same sign:

x	$(-\infty, 2)$	$(2, 3)$	$(3, +\infty)$
Numerator	$-$	$-$	$-$
Denominator	$+$	$-$	$+$
Fraction	$-$	$+$	$-$

Finally we have:

$$\boxed{2 < x < 3}$$

Ex d) Absolute value

Solve the following inequality:

$$\frac{|x|}{x-1} > \frac{x+1}{2x-1}$$

Domain:

$$D = x \in \mathbb{R} \setminus \left\{ \frac{1}{2}, 1 \right\}$$

Now recall the abs value definition:

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

In this case $f(x) = x$ so we have:

$$\begin{cases} \dfrac{x}{x-1} > \dfrac{x+1}{2x-1} & x \geq 0 \\ \dfrac{-x}{x-1} > \dfrac{x+1}{2x-1} & x < 0 \end{cases}$$

Remeber:

- The solution to each part of the system must be put in intersection with the condition
- The solutions of the parts of a system are in union with each other

Start with case 1:
Rewrite so that it is a single fraction:

$$\frac{x}{x-1} - \frac{x+1}{2x-1} > 0 \implies \frac{x(2x-1) - (x+1)(x-1)}{(x-1)(2x-1)} > 0$$

- Numerator:

$$x(2x-1) - (x^2-1) = 2x^2 - x - x^2 + 1 = x^2 - x + 1$$

Find the zeroes of the numerator:

$$x^2 - x + 1 = 0 \to \Delta = -3$$

Since $\Delta < 0$ and $a = 1 > 0$, the numerator is strictly positive.

- Denominator:

$$(x-1)(2x-1) > 0 \iff x < \frac{1}{2} \cup x > 1$$

Sign analysis for Case 1:

x	$\left(-\infty, \frac{1}{2}\right)$	$\left(\frac{1}{2}, 1\right)$	$(1, +\infty)$
Numerator	$+$	$+$	$+$
Denominator	$+$	$-$	$+$
Fraction	$+$	$-$	$+$

So the fraction is > 0 on $\left(-\infty, \frac{1}{2}\right) \cup (1, +\infty)$.

Intersecting with the case condition $x \geq 0$:

$$S_1 = \left[0, \frac{1}{2}\right) \cup (1, +\infty)$$

Case 2:
Same procedure as case 1:

$$\frac{-x}{x-1} - \frac{x+1}{2x-1} > 0 \implies \frac{-x(2x-1) - (x+1)(x-1)}{(x-1)(2x-1)} > 0$$

- Numerator:

$$-2x^2 + x - (x^2-1) = -3x^2 + x + 1$$

Find the zeroes of the numerator:

$$x_{1,2} = \frac{-1 \pm \sqrt{13}}{-6} = \frac{1 \mp \sqrt{13}}{6}$$

Since $a = -3 < 0$, the parabola opens downward:

$$N > 0 \iff x_1 = \frac{1 - \sqrt{13}}{6} < x < \frac{1 + \sqrt{13}}{6} = x_2$$

Denominator:
The denominator roots are $\frac{1}{2} = 0.5$ and 1.

Sign analysis for the factors:

x	$(-\infty, x_1)$	$(x_1, 1/2)$	$(1/2, x_2)$	$(x_2, 1)$	$(1, +\infty)$
Numerator	$-$	$+$	$+$	$-$	$-$
Denominator	$+$	$+$	$-$	$-$	$+$
Fraction	$-$	$+$	$-$	$+$	$-$

The fraction is > 0 on $\left(x_1, \frac{1}{2}\right) \cup (x_2, 1)$.

Intersecting with the case condition $x < 0$:

$$S_2 = \left(\frac{1-\sqrt{13}}{6}, 0\right)$$

Taking the union of both cases $S = S_1 \cup S_2$:

$$S = \left(\frac{1-\sqrt{13}}{6}, 0\right) \cup \left[0, \frac{1}{2}\right) \cup (1, +\infty)$$

Notice that 0 is included in the solution set, so the first two intervals join together:

$$\boxed{S = \left(\frac{1-\sqrt{13}}{6}, \frac{1}{2}\right) \cup (1, +\infty)}$$

Ex f) Square Root

Solve the following inequality:

$$\sqrt{x^2 - 6x} > x + 2$$

First notice that the domain is

$$x^2 - 6x \geq 0 \rightarrow D = (-\infty, 0] \cup [6, +\infty)$$

First notice that when $x + 2 < 0$ the solution inequality holds (in the domain). So one part of the solution is:

$$S_1 = D \cap \{x < -2\} = \{x < -2\}$$

Now for $x + 2 \geq 0$ of both sides can be squared:

$$x^2 - 6x > x^2 + 4x + 4 \rightarrow 0 > 10x + 4 \rightarrow x < -\frac{2}{5}$$

This clearly for the values bigger than -2 in the domain, that is

$$S_2 = D \cap \left\{\{x \geq -2\} \cap \left\{x < -\frac{2}{5}\right\}\right\} = D \cap \left\{-2 \leq x < -\frac{2}{5}\right\} = \left\{-2 \leq x < -\frac{2}{5}\right\}$$

So the result is:

$$S = S_1 \cup S_2 = \{x < -2\} \cup \left\{-2 \leq x < -\frac{2}{5}\right\} = x < -\frac{2}{5}$$

Ex i) Mixed Together

Solve the following inequality:

$$\sqrt{|x^2-4|}-x\geq 0$$

Domain: $|x^2-4|\geq 0 \rightarrow D=\mathbb{R}$

Now recall the abs value definition:

$$|f(x)|=\begin{cases} f(x) & f(x)\geq 0 \\ -f(x) & f(x)<0 \end{cases}$$

In this case $f(x)=x^2-4=(x+2)(x-2)$ so $f(x)\geq 0 \rightarrow x\leq -2 \cup x\geq 2$

Now we can rewrite the inequality as a system of two inequalities:

$$\sqrt{|x^2-4|}-x\geq 0 \rightarrow \begin{cases} \sqrt{x^2-4}-x\geq 0 & x\leq -2 \cup x\geq 2 \\ \sqrt{4-x^2}-x\geq 0 & -2<x<2 \end{cases}$$

Each system can be solved independently from the other.
Call the conditions of the cases D_1 and D_2 respectively.

Solve first case:
Recall that when RHS (x) is negative there is always a solution.

$$\sqrt{x^2-4}\geq x$$

Therefore one part of the solution includes $x<0$

Now study in the interval $x\geq 0$: since we are sure that both sides are ≥ 0 we can square both sides:

$$x^2-4\geq x^2 \rightarrow -4\geq 0$$

Not possible, so the only solution to this part is when x is negative:

$$S_1=x\leq 0 \cap D_1=x\leq -2$$

Solve the second equation:
As before a solution always exists for $x<0$. Now consider only the positive values ($x\geq 0$) in order to be able to square both sides:

$$\sqrt{4-x^2}\geq x \rightarrow 4-x^2\geq x^2 \rightarrow 4-2x^2\geq 0 \rightarrow -\sqrt{2}\leq x\leq \sqrt{2}$$

So we have $S_2=D_2\cap\left\{x<0\cup -\sqrt{2}\leq x\leq \sqrt{2}\right\}=D_2\cap x\leq \sqrt{2}=-2<x\leq \sqrt{2}$

The result is therefore the union of both cases which yields:

$$\boxed{S=x\leq -2\cup -2<x\leq \sqrt{2}\rightarrow S=x\leq \sqrt{2}}$$

1.2) Subsets of $\mathbb{R}$

Ex a)

Describe the following set:

$$A=S_1\cap S_2=\left\{x\in\mathbb{R}:x^2+4x+13<0\right\}\cap\left\{x\in\mathbb{R}:3x^2+5>0\right\}$$

Subset 1:

$$x^2 + 4x + 13 < 0 \to \Delta = -36 \implies S_1 = \emptyset$$

any intersection with an empty set is an empty set:

$$\boxed{A = \emptyset \cap S_2 = \emptyset}$$

In general recall:

$$\emptyset \cap S = \emptyset \qquad \emptyset \cup S = S$$

Ex c)

Describe the following set:

$$C = \left\{ x \in \mathbb{R} : \frac{x^2 - 5x + 4}{x^2 - 9} < 0 \right\} \cup \left\{ x \in \mathbb{R} : x + \sqrt{7x + 1} = 17 \right\}$$

Subset 1:
First notice that the domain is $x \neq \pm 3$

Numerator:

$$x^2 - 5x + 4 = 0 \to x_{1,2} = 1,\ 4 \to N > 0 \iff x < 1 \cup x > 4$$

Denominator:

$$x^2 - 9 = (x + 3)(x - 3) = 0 \to D > 0 \iff x < -3 \cup x > 3$$

x	$(-\infty, -3)$	$(-3, 1)$	$(1, 3)$	$(3, 4)$	$(4, +\infty)$
Numerator	$+$	$+$	$-$	$-$	$+$
Denominator	$+$	$-$	$-$	$+$	$+$
Fraction	$+$	$-$	$+$	$-$	$+$

so $S_1 = (-3, 1) \cup (3, 4)$

Subset 2:
This is just an equality:

$$\sqrt{7x + 1} = 17 - x$$

The domain is $7x + 1 \geq 0 \to D = \left\{ x \geq -\frac{1}{7} \right\}$
Moreover as before, both sides must have the same sign when squaring and since the sqrt is always ≥ 0 we must have that also $17 - x \geq 0 \to x \leq 17$ so we can further reduce the domain to

$$D = \left[-\frac{1}{7}, 17 \right]$$

Square both sides:

$$7x + 1 = (17 - x)^2 \to 7x + 1 = x^2 - 34x + 17^2 \to x^2 - 41x + 288 = 0$$

$$x_{1,2} = 9,\ 32$$

But only 9 is acceptable as $32 \notin D$

Finally:

$$\boxed{C = (-3, 1) \cup (3, 4) \cup \{9\}}$$

1.3) Max/Min, Sup/Inf

Recall the theory

Definition (Upper/Lower Bound)**.**

$M \in \mathbb{Q}$ is called **upper bound** for A if $M \geq a, \forall a \in A$.

Respectively it is called **lower bound** if $M \leq a, \forall a \in A$

Notice M does not need to be in A
Example: consider set $A = [0, 1)$ any $m \leq 0$ is a lower bound and any $M \geq 1$ is upper bound

Definition (Maximum/Minimum of A)**.**

If $\overline{M}$ is an upper bound for A and $\overline{M} \in A$, then $\overline{M}$ is the *minimum* possible *upper* bound and is the **maximum** (of A).

If $\overline{m}$ is an lower bound for A and $\overline{m} \in A$, then $\overline{m}$ is the *maximum* possible *lower* bound and the **minimum** (of A).

In this case $\overline{M} \in A$ by definition.
From the previous example: $\overline{m} = \max\{m \leq 0\} = 0 \in A$ is the correct minimum of A, however $\overline{M} = \min\{M \geq 1\} = 1 \notin A$ means that we don't have a maximum

Definition (Supremum/Infimum)**.**

The **supremum of** A $\sup A$ is the minimum of the upper bounds of A.

The **infimum of** A $\inf A$ is the maximum of the lower bounds of A.

Finish the example: we know $\overline{m} = 0 \in A$ so this is both minimum and infimum of A. We know also $\overline{M} = 1 \notin A$, therefore $\sup \mathrm{A} = 1$ but it is not the maximum.

Recall that if a set is unbounded, then

$$\sup A = +\infty \quad \text{if } A \text{ is unbounded from above}$$

$$\inf A = -\infty \quad \text{if } A \text{ is unbounded from below}$$

Ex a)

Tell wether this subsets is bounded and the sup/inf and if it allow for maximum/minimum.

$$A = \left\{ x \in \mathbb{R} : x = n \text{ or } x = \frac{1}{n^2}, n \in N \setminus \{0\} \right\}$$

It is immediately clear that A is the union of two sets, mainly

$$A = S_1 \cup S_2 \text{ where } \begin{array}{l} S_1 = \{n \in \mathbb{N} \setminus \{0\}\} = \{1, 2, 3, 4, \ldots\} \\ S_2 = \left\{ \frac{1}{n^2} \mid n \in \mathbb{N} \setminus \{0\} \right\} = \left\{ 1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \ldots \right\} \end{array}$$

Find the bounds on both sets individually (not required but the result is clearer).

Study the first subset S_1: notice that it is just the natural numbers with zero excluded, so its properties are already known.

- No upper bound, so also no max
  - Supremum: since unbounded; sup $S_1 = +\infty$
- Lower bound: $m_1 \leq 1$
  - Minimum: $\overline{m}_1 = \max m_1 \leq 1 = 1 \in S$

Study the second subset S_2: Intuitively we know that the function describing the set is monotonically decreasing: we can express it by saying:

Call x_n the value of x for any chosen n, then $x_{n+1} < x_n$.
This is easily shown as the following inequality is true for $n \in \mathbb{N} \setminus \{0\}$ (check at home):

$$\frac{1}{(n+1)^2} < \frac{1}{n^2}$$

Therefore for every number $x_n > 0$ we can find a smaller number such that $0 < x_{n+1} < x_n$

Is 0 the maximum lower bound?
By contradiction suppose there exists a lower bound $\delta > 0$ then if 0 is the maximum lower bound we have

$$\exists n \in \mathbb{N} \setminus \{0\} \quad \text{such that} \quad \frac{1}{n^2} < \delta$$

Now compute

$$\frac{1}{n^2} < \delta \to n^2 > \frac{1}{\delta} \to n > \frac{1}{\sqrt{\delta}}$$

Because $\delta > 0$, the number $\frac{1}{\sqrt{\delta}}$ is just a fixed positive real number and an appropriate n exists.

With this digression aside, we also know that $0 \notin S_2$.
As before analyze:

- Lower bound: $m_2 \leq 0$
  - No minimum: $\overline{m}_2 = \max m_2 \leq 0 = 0 \notin S_2$
  - Infimum is $\inf S_2 = 0$
- Upper bound: $M_2 \geq 1$
  - $\overline{M}_2 = \min M_2 \geq 1 = 1 \in S_2$

Putting all together:
Now consider the union, so:

- No upper bound since S_1 is unbounded
  - therefore sup $A = +\infty$

- Lower bound: The infimum is the **maximum** of all lower bounds: $\inf A = \min(\inf S_1, \inf S_2) = \min(1, 0) = 0$. Since $0 \notin A$, $\min A$ does not exist.

2) Lesson 16/10/26

2.1) Domains

Recall:

$$f(x) = \begin{matrix} \dfrac{a(x)}{g(x)} & \text{dom } f = \{g(x) \neq 0\} \\ \sqrt{g(x)} & \text{dom } f = \{g(x) \geq 0\} \\ \log(g(x)) & \text{dom } f = \{g(x) > 0\} \end{matrix}$$

a) Fraction and Sqrt

Determine the domains of the following function:

$$f(x) = \frac{\sqrt{x^2 - 3x - 4}}{x + 5}$$

Here the domain is given by two solutions that must hold simultaneously.

Fraction:
Here $D_1 = \{x + 5 \neq 0\} = \{x \neq -5\}$

Sqrt:
Here $D_2 = \{x^2 - 3x - 4 \geq 0\}$. The zeroes of this function are $x_{1,2} = -1, 4$ so the inequality holds for $D_2 = \{x \leq -1 \cup x \geq 4\}$

Finally

$$\text{dom } f = D_1 \cap D_2 = \{x \leq -1 \cup x \geq 4\} \setminus \{-5\} = (-\infty, -5) \cup (-5, -1] \cup [4, +\infty)$$

where the last part is the correct formal way to write the result.

b) Logarithm

Determine the domains of the following function:

$$f(x) = \log(x^2 - x)$$

The domain is given by $x^2 - x > 0$, so solve the inequality:

$$x^2 - x = x(x - 1) > 0$$

The first term is positive for $x > 0$, the second for $x > 1$, so

$$\text{dom } f = \{x < 0 \cup x > 1\}$$

c) System

Determine the domains of the following function:

$$f(x) = \begin{cases} \dfrac{1}{2x + 1} & x \geq 0 \\ e^{\sqrt{x+1}} & x < 0 \end{cases}$$

Here the domain is given by the union of the two cases.

Case 1:
Here the domain is given by $D_1^* = \{2x + 1 \neq 0\} = \left\{x \neq -\frac{1}{2}\right\}$. However since this is a system, this must hold

only inside the specified case, so $D_1 = D_1^* \cap \{x \geq 0\} = \{x \geq 0\}$ since $-\frac{1}{2} \notin \{x \geq 0\}$, so this part of the map is always valid.

Case 2:
Here the domain is given by $x + 1 \geq 0 \rightarrow x \geq -1$ which combined with the system is $x \in [-1, 0)$

The final domain becomes the union of the two cases, so

$$\text{dom } f = \{x \geq 0\} \cup \{-1 \leq x < 0\} = \{x \geq -1\}$$

2.2) Symmetries and Periodicity

If a function is symmetric, it can have either even or odd symmetry:

- Even Symmetry: mirrored along y axis: $f(x) = f(-x)$
- Odd Symmetry: mirrored along x and y axis: $f(x) = -f(-x)$

a) Sqrt and Abs

Sketch the graph of the following function indicating the possible symmetries and/or periodicity:

$$f(x) = \sqrt{1 - |x|}$$

Domain:

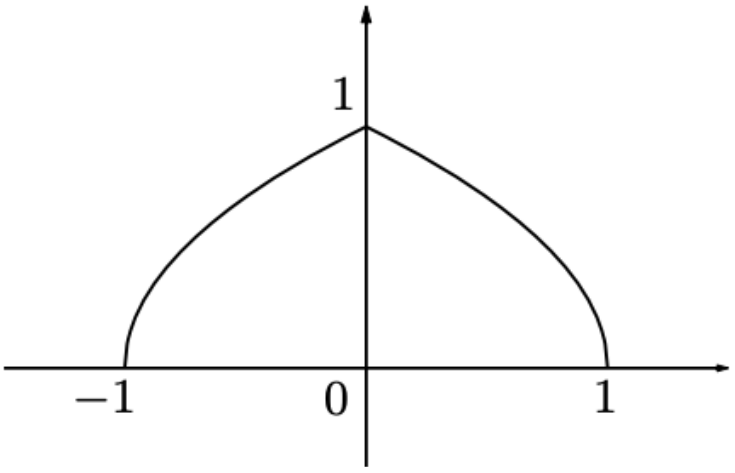
$$D = \{1 - |x| \geq 0\} = \{|x| \leq 1\} = \{x \in [-1, 1]\}$$

So clearly the function cannot be periodic. By definition, a periodic function requires $x \in \text{dom } f \implies x + T \in \text{dom } f$ for all x, which cannot hold on a bounded interval

Symmetry:
Since f is the result of a sqrt, it can only be non negative (roots at $x = \pm 1$), therefore an odd symmetry is not possible (only possible if f is constant 0). Try the even symmetry:

$$f(-x) = \sqrt{1 - |-x|} = \sqrt{1 - |-1| \cdot |x|} = \sqrt{1 - 1 \cdot |x|} = f(x)$$

The final sketch can be done by sampling points only between $[0, 1]$ since then it can be mirrored.



Sketch

b) Cosine

Sketch the graph of the following function indicating the possible symmetries and/or periodicity:

$$f(x) = 1 + \cos(2x)$$

There is no restriction in the domain, so start with the symmetries. Again since $\cos(z) \in [-1, 1]$ the function returns only non-negative values, so an odd symmetry is excluded, moreover since $\cos(z)$ is an even function, we can guess that the function is even, let's prove it:

$$f(-x) = 1 + \cos(-2x) = 1 + \cos(2x) = f(x)$$

Periodicity:
T_1: 1 is a constant function. A constant function $c(x) = c$ satisfies $c(x + T) = c(x)$ for **every** $T > 0$. Therefore, every positive real number is a period. However it does not have a fundamental period (0 cannot be period)
T_2: $\cos(\theta)$ has period 2π, that is $\cos(\theta + 2\pi) = \cos(\theta)$, but what about $g(x) = \cos(2x)$? Let's try the generic case $\cos(ax)$, $a > 0$

By the definition of period, we must find a $T > 0$ such that

$$\min_T \{T > 0 : g(x + T) = g(x)\ \forall x \in \mathbb{R}\} = \min_T \{\cos(a(x + T)) = \cos(ax)\}$$

Recall that $\cos(\theta + 2\pi) = \cos(\theta)$ and by comparing it with the argument of the minimum we get

$$aT = 2\pi \to T = \frac{2\pi}{a}$$

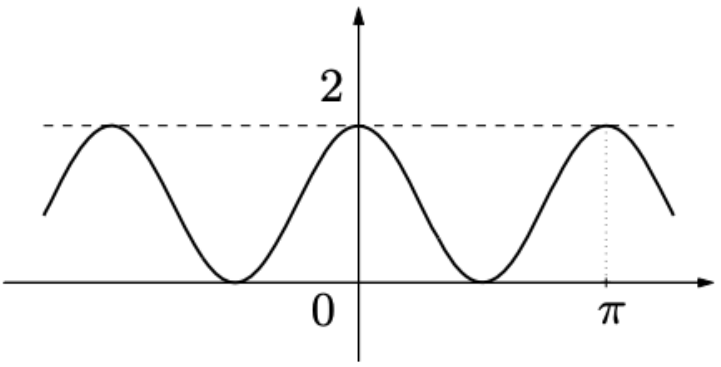
And in our case $T_2 = \pi$

Adding a constant is merely a vertical translation and leaves the periodic properties unchanged:

$$f(x + T) = f(x) \iff 1 + \cos(2(x + T)) = 1 + \cos(2x) \iff \cos(2(x + T)) = \cos(2x)$$

Therefore, the fundamental period of $f(x)$ is simply the fundamental period of $\cos(2x)$:

$$T = \frac{2\pi}{2} = \pi$$



Sketch

Two sinusoids

Determine the period of the following function:

$$f(x) = \sin(2x) + \cos(\pi x)$$

From the previous exercise we know that a sinusoidal has period

$$T = \frac{2\pi}{a}$$

So we know the two periods:

$$T_1 = \frac{2\pi}{2} = \pi \qquad T_2 = \frac{2\pi}{\pi} = 2$$

Now for the final period, recall that **the sum of two periodic functions is periodic if and only if the ratio of their periods is a rational number.**

Check the ratio:

$$\frac{T_1}{T_2} = \frac{\pi}{2} \notin \mathbb{Q}$$

so the function is aperiodic.

Now consider

$$f(x) = \sin(2x) + \cos(3x)$$

therefore

$$\begin{aligned} T_1 &= \pi \\ T_2 &= \frac{2\pi}{3} \end{aligned} \qquad \frac{T_1}{T_2} = \frac{3}{2} \in \mathbb{Q}$$

We know the function is periodic! But how is the period found?
To find the fundamental period T, any common period must satisfy $f(x+T) = f(x)$ for all x. For the sum of two independent sinusoids, this requires T to be a simultaneous integer multiple of both individual periods T_1 and T_2:

$$T = k_1 T_1 = k_2 T_2 \quad \text{for some } k_1, k_2 \in \mathbb{Z}^+$$

That is:

$$k_1 \pi = k_2 \left(\frac{2\pi}{3}\right) \implies k_1 = \frac{2}{3} k_2 \implies \frac{k_1}{k_2} = \frac{2}{3} \qquad k_1 = 2, k_2 = 3$$

and thus the period is

$$T = 2\pi = 3 \cdot \frac{2\pi}{3}$$

1) Arcsin

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \arcsin\left(\left|x^3 - \frac{1}{2}\right|\right)$$

Properties of $\arcsin(t)$
The sin function is not bijective (surjective but not injective) the map is $\mathbb{R} \to [-1, 1]$. So the arcsin is defined on the restriction in which this function is strictly increasing and bijective, that is

$$\sin\big|_{[-\frac{\pi}{2}, \frac{\pi}{2}]} : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \longrightarrow [-1, 1] \implies \arcsin : [-1, 1] \longrightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

The domain is therefore

$$\text{dom } \arcsin = [-1, 1]$$

The sign coincides with the sign of the argument.

Domain:
Recall that $\arcsin(t)$ is defined for $t \in [-1, 1]$, so we must impose this as the domain:

$$-1 \leq \left|x^3 - \frac{1}{2}\right| \leq 1$$

Since the absolute value is always non-negative we have

$$\left|x^3 - \frac{1}{2}\right| \leq 1 \to -1 \leq x^3 - \frac{1}{2} \leq 1 \to -\frac{1}{\sqrt[3]{2}} \leq x \leq \sqrt[3]{\frac{3}{2}}$$

So we have

$$\operatorname{dom} f = \left[-\frac{1}{\sqrt[3]{2}}, \sqrt[3]{\frac{3}{2}} \right]$$

Here we can notice that since the domain is not symmetric wrt the origin, the function does not have any symmetries, that is: A function can have symmetry **only if** its domain is symmetric with respect to the origin

$$x \in \operatorname{dom} f \implies -x \in \operatorname{dom} f$$

Moreover, since it has a finite domain it cannot be periodic.

Sign:
In our case since we have only positive values, the function will always be positive

$$\arcsin(t) \in \left[0, \frac{\pi}{2}\right] \implies \arcsin(t) \geq 0$$

The root is at $\left|x^3 - \frac{1}{2}\right| = 0 \to x = \sqrt[3]{\frac{1}{2}}$

3) Exponential

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \log(e^{2x} - 4e^x + 4)$$

First we do a substitution, $y \triangleq e^x$ and therefore

$$f(y) = \log(y^2 - 4y + 4)$$

Domain:
From the log we impose $y^2 - 4y + 4 > 0$
The roots are $y_{1,2} = 2$ which for x becomes:

$$\begin{aligned} 2 &= e^x \\ \log 2 &= \log e^x \\ x &= \log 2 \end{aligned}$$

so the domain is just dom $f = \mathbb{R} \setminus \{\log 2\} = (-\infty, \log 2) \cup (\log 2, +\infty)$
Since domain is not symmetric wrt to origin, we don't have any symmetry.

Sign:
Recall $\log(z) \geq 0 \iff z \geq 1$, so

$$y^2 - 4y + 4 \geq 1 \to y^2 - 4y + 3 \geq 0$$

The roots are $y_{1,2} = 1, 3 \to x_{1,2} = 0, \log 3$
The inequality holds for $x \leq 0 \cup x \geq \log 3$

So $f(x) \geq 0 \iff x \leq 0$ or $x \geq \log 3$
And the roots are at $x = 0,\ x = \log 3$

7) Hyperbolic Cosine

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \frac{1}{\cosh(\sin(x))}$$

What is the hyperbolic cosine?

$$\cosh(t) = \frac{e^t + e^{-t}}{2}$$

So this is always defined and is always positive.
This is also an even function:

$$\cosh(-t) = \frac{e^{-t} + e^t}{2} = \cosh(t)$$

Domain:
First define $a = \sin(x) \in [-1, 1]$
The denominator cannot be 0 so

$$\cosh(a) = \frac{e^a + e^{-a}}{2} \neq 0$$

Which is valid $\forall a$ as $e^k > 0 \quad \forall k \in \mathbb{R}$.

So

$$\text{dom } f = \mathbb{R}$$

Sign:
Hyperbolic cosine is always positive so also this function is always positive with no zero

Symmetry:
Cosh is even $\cosh(x) = \cosh(-x)$, sin is odd $\sin(-x) = -\sin(x)$.
Try even:

$$f(-x) = \frac{1}{\cosh(\sin(-x))} = \frac{1}{\cosh(-\sin(x))} = \frac{1}{\cosh(\sin(x))} = f(x)$$

So it is even!

Is it periodic?
As before impose

$$f(x + T) = f(x) \quad \text{for all } x \in \mathbb{R}$$

Substituting

$$\frac{1}{\cosh(\sin(x + T))} = \frac{1}{\cosh(\sin(x))} \to \cosh(\sin(x + T)) = \cosh(\sin(x))$$

Easy way: Try period of $\sin(x)$ (2π):

$$\begin{aligned} \cosh(\sin(x + 2\pi)) &= \cosh(\sin(x)) \\ \cosh(\sin(x)) &= \cosh(\sin(x)) \end{aligned}$$

Harder way:
Since cosh has even symmetry:

$$\cosh(t) = \cosh(-t) \implies \cosh(t) = \cosh(|t|) \quad \forall t \in \mathbb{R}$$

And therefore (prove at home):

$$\cosh(|u|) = \cosh(|v|) \iff |u| = |v|$$

And thus impose:

$$|\sin(x + T)| = |\sin(x)| \quad \forall x \in \mathbb{R}$$

Squaring both sides:

$$\sin^2(x+T) = \sin^2(x)$$

Using the identity $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$:

$$\frac{1-\cos(2x+2T)}{2} = \frac{1-\cos(2x)}{2} \iff \cos(2x+2T) = \cos(2x) \quad \forall x \in \mathbb{R}$$

Therefore the period is the same as $\cos(2x)$ which is $T = \pi$ and thus 2π is not the fundamental period (it is a period nonetheless).

10)

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \frac{\log(\sin(x))}{\sin(x) - 1}$$

Domain:
Denominator forces $\sin(x) \neq 1$ which holds for $x \neq \frac{\pi}{2} + 2k\pi \quad k \in \mathbb{Z}$
The log forces $\sin(x) > 0$ which is true for $x \in (2k\pi, \pi + 2k\pi) \quad k \in \mathbb{Z}$
so the generic k-th interval is $(2k\pi, \frac{\pi}{2} + 2k\pi) \cup (\frac{\pi}{2} + 2k\pi, \pi + 2k\pi)$
The domain is the union of all these intervals:

$$\text{dom } f = \bigcup_{k \in \mathbb{Z}} (2k\pi, \frac{\pi}{2} + 2k\pi) \cup (\frac{\pi}{2} + 2k\pi, \pi + 2k\pi)$$

which is not symmetric wrt the origin and thus there is no symmetry

Sign:
The denominator is always negative since $\max \sin(x) = 1$ and thus $\max \sin(x) - 1 = 0$ (which is excluded by domain)
The numerator is also always negative:

$$\log(\sin(x)) \geq 0 \iff \sin(x) \geq 1 \iff \sin(x) = 1$$

which is excluded from the domain

Therefore this function is always positive

Periodicity:
Notice that the domain itself needs to be
periodic (or to be $\mathbb{R}$). In this case the domain is given by intervals given by $\sin(x)$ and it can be shifted only by $2k\pi$ to hold correctly. Therefore we can directly try with $T = 2\pi$:

$$f(x+2\pi) = \frac{\log(\sin(x+2\pi))}{\sin(x+2\pi) - 1} = \frac{\log(\sin(x))}{\sin(x) - 1} = f(x)$$

3) Lesson 23/10/26

3.1) Limits

Recall the following indeterminate forms:

$$(+\infty) + (-\infty) \tag{1}$$
$$0^{\pm} \cdot (\pm\infty) \tag{2}$$
$$\frac{0^{\pm}}{0^{\pm}} \tag{3}$$
$$\frac{\pm\infty}{\pm\infty} \tag{4}$$

Where D' is the
Limits at infinity are used to find asymptote. This has an equation of form:

$$y = mx + q$$

Two cases happen:

- $\lim_{x\to\infty} f(x) = c \in \mathbb{R} \longrightarrow$ asymptote is $y = c$ $(m = 0)$
- $\lim_{x\to\infty} f(x) = \pm\infty \longrightarrow$ here we might have an oblique asymptote $(m \neq 0)$ or a generic divergence. The slope can be found with:

  $$m = \lim_{x\to\pm\infty} \frac{f(x)}{x}$$

  if $m = \pm\infty$ there is no asymptote, otherwise q can be found as

  $$q = \lim f(x) - mx$$

1) Fraction

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \frac{x-5}{x^2-9}$$

Domain:

$$D = x \in \mathbb{R} \setminus \{\pm 3\}$$

Sign:

- Numerator $> 0 \longrightarrow x > 5$
- Denominator $> 0 \longrightarrow x \in (-\infty, -3) \cup (3, +\infty)$

$$f(x) > 0 \iff x \in (-3, 3) \cup (5, +\infty)$$

Limits at infinity:

- Positive Infinity:

  $$\lim_{x\to\infty} f(x) = \frac{\infty}{\infty} \quad \text{indeterminate form !}$$

  Notice that:

$$f(x) = \frac{x(1-\frac{5}{x})}{x^2(1-\frac{9}{x^2})} = \frac{(1-\frac{5}{x})}{x(1-\frac{9}{x^2})}$$

The fractional terms tend to zero ($\lim_{x\to\infty} c/x^n = 0$) so

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{(1-\frac{5}{x})}{x(1-\frac{9}{x^2})} = \frac{1-0^+}{\infty(1-0^+)} = \frac{1^-}{\infty} = 0^+$$

- Negative Infinity:

$$\lim_{x\to-\infty} f(x) = \frac{1-0^-}{-\infty(1-0^-)} = \frac{1^+}{-\infty} = 0^-$$

Limits at accumulation points:
Here the use of the sign will be useful to determine if we have 0^+ or 0^- at the denominator

- At $+3$:

$$\lim_{x\to3^+} f(x) = \frac{-2}{0^+} = -\infty$$

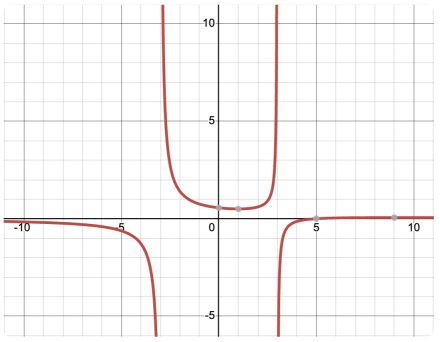
check also 3^-:

$$\lim_{x\to3^-} f(x) = \frac{-2}{0^-} = +\infty \neq \lim_{x\to3^+} f(x)$$

So the limit doesn't exist.

- At -3:

$$\lim_{x\to-3^+} f(x) = \frac{-8}{0^-} = +\infty \neq \lim_{x\to-3^-} f(x) = \frac{-8}{0^+} = -\infty$$



Graph

2) Oblique Asymptote

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \frac{x^3}{x^2-1}$$

Domain:

$$D = x \in \mathbb{R} \setminus \{\pm1\}$$

Sign:

- Numerator $> 0 \longrightarrow x > 0$
- Denominator $> 0 \longrightarrow x \in (-\infty,-1) \cup (1,+\infty)$

$$f(x) > 0 \iff x \in (-1,0) \cup (1,+\infty)$$

Limits at infinity:

- Positive Infinity:

$$\lim_{x\to\infty} f(x) = \frac{\infty}{\infty} \quad \text{indeterminate form !}$$

Notice that:

$$f(x) = \frac{x^3}{x^2(1-\frac{1}{x^2})} = \frac{x}{1-\frac{1}{x^2}}$$

The fractional terms tend to zero ($\lim_{x\to\infty} c/x^n = 0$) so

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{x}{1-\frac{1}{x^2}} = \frac{\infty}{1-0^+} = \infty$$

Is there an oblique asymptote?

$$m = \lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} \frac{1}{1-\frac{1}{x^2}} = 1$$
$$q = \lim_{x\to\infty} f(x) - x = \lim_{x\to\infty} \frac{1}{x(1-\frac{1}{x^2})} = 0^+$$

- Negative Infinity:

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{x}{1-\frac{1}{x^2}} = \frac{\infty}{1-0^+} = -\infty$$

Is there an oblique asymptote?

$$m = \lim_{x\to-\infty} \frac{f(x)}{x} = \lim_{x\to-\infty} \frac{1}{1-\frac{1}{x^2}} = 1$$
$$q = \lim_{x\to-\infty} f(x) - x = \lim_{x\to-\infty} \frac{1}{x(1-\frac{1}{x^2})} = 0^-$$

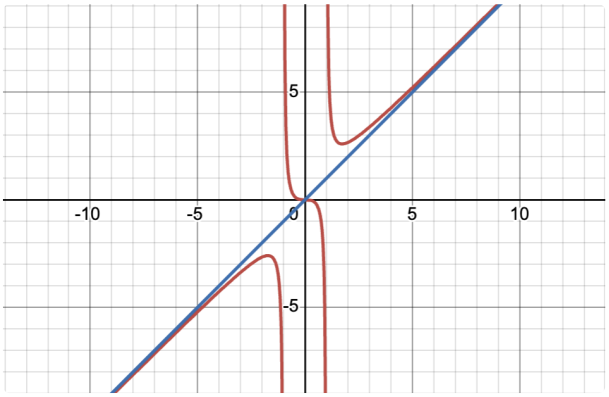
Limits at accumulation points:

- At $+1$:

$$\lim_{x\to1^+} f(x) = \frac{1}{0^+} = \infty \neq \lim_{x\to1^-} f(x) = \frac{1}{0^-} = -\infty$$

- At -1:

$$\lim_{x\to-1^+} f(x) = \frac{-1}{0^-} = +\infty \neq \lim_{x\to-1^-} f(x) = \frac{-1}{0^+} = -\infty$$



4) Log

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \log\left(\frac{x+1}{x-1}\right)$$

Domain:

$$D = \left\{\frac{x+1}{x-1} > 0\right\} \cap \{x \neq 1\} = (-\infty, -1) \cup (1, +\infty)$$

Sign:

$$f(x) > 0 \iff \frac{x+1}{x-1} > 1 \to x > 1$$

We don't have any zero.

Limits at infinity:

- Positive Infinity:

  $$\lim_{x\to\infty} f(x) = \log\left(\frac{\infty}{\infty}\right) \quad \text{indeterminate form!}$$

  Notice that $\frac{x+1}{x-1} = \frac{x(1+1/x)}{x(1-1/x)} = \frac{1+\frac{1}{x}}{1-\frac{1}{x}}$ so

  $$\lim_{x\to\infty} f(x) = \log(1^+) = 0^+$$

  where we know that the solution must be positive from the domain.
  In general recall that a ratio of polynomial can be solved by just looking at the max degree of numerator and denominator!
- Negative Infinity:

  $$\lim_{x\to-\infty} f(x) = \log\left(1^-\right) = 0^-$$

  sign is deduced from previous sign analysis!

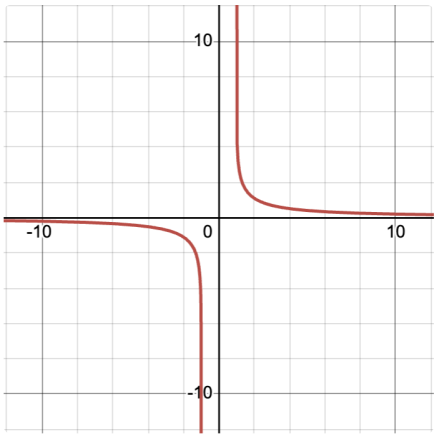
Limits at accumulation points:

- At 1 (only 1^+ due to domain):

  $$\lim_{x\to 1^+} f(x) = \log\left(\frac{2}{0^+}\right) = \log(+\infty) = \infty$$

- At -1 (only -1^- due to domain):

  $$\lim_{x\to -1^-} f(x) = \log\left(\frac{0^-}{-2}\right) = \log(0^+) = -\infty$$



Graph

7) Log o Log

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \frac{\log(\log(x))}{1+\log(x)}$$

Domain:
We have three conditions:

- Inner logarithm requires: $x > 0$
- Outer logarithm in the numerator requires: $\log(x) > 0 \implies x > 1$
- Denominator must not vanish: $1 + \log(x) \neq 0 \implies \log(x) \neq -1 \implies x \neq \frac{1}{e}$
  Since $\frac{1}{e} \approx 0.37 < 1$, the condition $x \neq \frac{1}{e}$ is automatically satisfied for $x > 1$.

$$D = (1, +\infty)$$

Sign:

- Denominator: For all $x \in D = (1, +\infty)$, $\log(x) > 0$, so $1 + \log(x) > 1 > 0$ (always positive).
- Numerator:

$$\log(\log(x)) > 0 \iff \log(x) > 1 \iff x > e$$

$$f(x) > 0 \iff x \in (e, +\infty)$$

Limits at infinity:

- Positive Infinity:

$$\lim_{x \to +\infty} f(x) = \frac{+\infty}{+\infty} \quad \text{indeterminate form!}$$

  Substitute $t = \log(x)$.
  As $x \to +\infty$, $t \to +\infty$:

$$\lim_{t \to +\infty} \frac{\log(t)}{1+t} = \lim_{t \to +\infty} \frac{\log(t)}{t\left(1+\frac{1}{t}\right)} = \lim_{t \to +\infty} \frac{\log(t)}{t} \cdot \frac{1}{1+\frac{1}{t}}$$

  Recall the hierarchy of infinities ($\lim_{t \to +\infty} \frac{\log(t)}{t} = 0^+$):

$$\lim_{x \to +\infty} f(x) = 0^+ \cdot 1 = 0^+$$

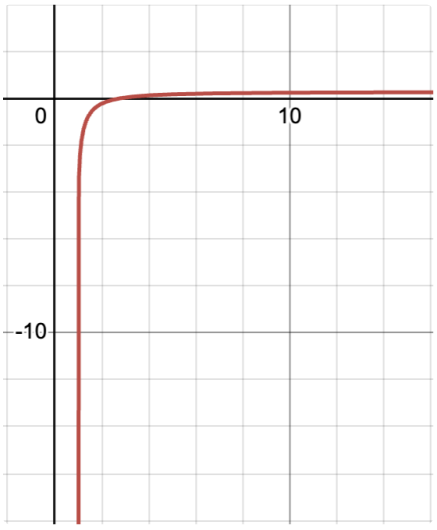
- Negative Infinity: Excluded by domain.

Limits at accumulation points:

- At 1 (only 1^+ due to domain):
  Numerator: as $x \to 1^+$, $\log(x) \to 0^+ \implies \log(\log(x)) \to -\infty$
  Denominator: $1 + \log(1^+) = 1 + 0^+ = 1^+$

$$\lim_{x \to 1^+} f(x) = \frac{-\infty}{1^+} = -\infty$$

  The line $x = 1$ is a vertical asymptote from the right.



Graph

8) Limited

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$$

Domain:

$$D = \{1 + x \geq 0\} \cap \{1 - x \geq 0\} \cap \{x \neq 0\} = [-1, 0) \cup (0, 1] = [-1, 1] \setminus \{0\}$$

Sign:
Two alternatives:

1. Find numerator denominator as usual (do at home):
   Numerator $> 0 \iff x \in (0, 1]$
   Denominator $> 0 \iff x > 0$
   so $f(x) > 0 \forall x \in D$
2. Rationalize

$$\begin{aligned} f(x) &= \frac{(\sqrt{1+x} - \sqrt{1-x})(\sqrt{1+x} + \sqrt{1-x})}{x(\sqrt{1+x} + \sqrt{1-x})} \\ &= \frac{(1+x) - (1-x)}{x(\sqrt{1+x} + \sqrt{1-x})} \\ &= \frac{2x}{x(\sqrt{1+x} + \sqrt{1-x})} \\ &= \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \quad (\text{ only since } D \text{ excludes } 0) \end{aligned}$$

Since $\sqrt{1+x}+\sqrt{1-x}>0$ for all $x \in [-1,1]$:

$$f(x) > 0 \quad \forall x \in D$$

We don't have any zeros.

Limits at infinity:
Positive and Negative Infinity are excluded by domain ($D \subset [-1,1]$, so $\pm\infty$ are not accumulation points of D).
There are no horizontal or oblique asymptotes.

Limits at accumulation points:

- At 0:

$$\lim_{x\to 0} f(x) = \frac{0}{0} \quad \text{indeterminate form!}$$

Using the rationalized form:

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{2}{\sqrt{1+x}+\sqrt{1-x}} = \frac{2}{\sqrt{1}+\sqrt{1}} = \frac{2}{2} = 1$$

Notice that:

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = 1$$

So the two-sided limit exists! $x=0$ is a removable discontinuity (eliminable singularity).

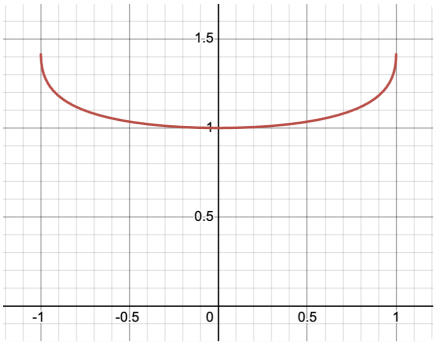
- At $+1$ (only 1^- due to domain): Since $1 \in D$, the function is continuous from the left:

$$\lim_{x\to 1^-} f(x) = f(1) = \frac{\sqrt{2}-0}{1} = \sqrt{2}$$

- At -1 (only $(-1)^+$ due to domain):Since $-1 \in D$, the function is continuous from the right:

$$\lim_{x\to -1^+} f(x) = f(-1) = \frac{0-\sqrt{2}}{-1} = \sqrt{2}$$

There are no vertical asymptotes.



Graph

4) Lesson 26/10/26

4.1) Standard Limits Exercises

3) Two Policemen (Harder!)

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \log(e^{2x} - 1)$$

Domain:

$$D = \{e^{2x} - 1 > 0\} = \{x > 0\}$$

Sign:

$$f(x) > 0 \iff e^{2x} - 1 > 1 \to x > \frac{\log 2}{2}$$

Limits at infinity:

- Positive Infinity:

  $$\lim_{x\to\infty} f(x) = \log(\infty - 1) = \infty$$

  Is there an oblique asymptote?
  Notice how we can rewrite the logarithm:

  $$f(x) = \log\left[e^{2x}\left(1 - \frac{1}{e^{2x}}\right)\right] = \log\left(e^{2x}\right) + \log\left(1 - e^{-2x}\right) = 2x + \log\left(1 - e^{-2x}\right)$$

  so we have an oblique asymptote:

  $$m = \lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} 2 + \frac{\log\left(1 - e^{-2x}\right)}{x} = 2 + \frac{\log(1^-)}{+\infty} = 2 + \frac{0^-}{+\infty} = 2$$

Alternatively the **squeeze theorem** can be used:

$$m = \lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} \frac{\log(e^{2x} - 1)}{x}$$

Now find the bounds:

- Upper bound: $e^{2x} - 1 < e^{2x} \to \log(e^{2x} - 1) < 2x$
- Lower bound: $e^{2x} - 1 > \frac{e^{2x}}{2} \to \log(e^{2x} - 1) > 2x - \log(2)$ valid for large x ($x > 1$)
  Now divide by $x > 0$

  $$2 - \frac{\log(2)}{x} \leq \frac{\log(e^{2x} - 1)}{x} \leq 2$$

  Take the limit:

  $$2 \leq \lim_{x\to\infty} \frac{f(x)}{x} \leq 2$$

  The interception is:

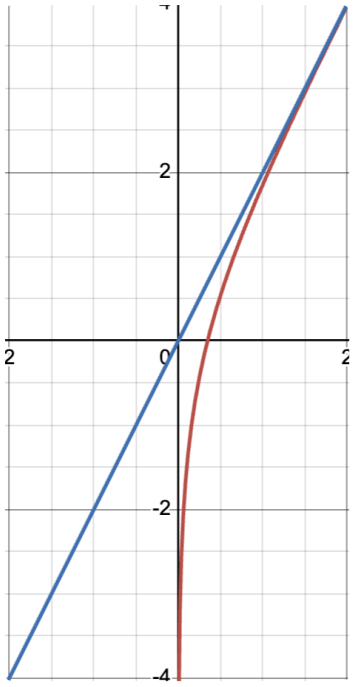
  $$\begin{aligned} q = \lim_{x\to\infty} f(x) - 2x &= +\infty - \infty \quad \text{indeterminate form!} \\ &= 2x + \log(1 - e^{-2x}) - 2x \\ &= \log(1 - e^{-2x}) = 0^- \end{aligned}$$

- Negative Infinity excluded by domain.

Limits at accumulation points:
Here we just need to check for 0^+ as 0^- is not in the domain!

- At 0^+:

$$\lim_{x \to 0^+} f(x) = \log(1^+ - 1) = \log(0^+) = -\infty$$



Asymptote in Blue

4.2) Taylor Expansion

A **Taylor expansion** takes a smooth function $f(x)$ near a specific point x_0 and replaces it with a **polynomial clone** $P_n(x)$ of degree n, plus a tiny error term (the remainder). This is useful as it transforms complicated functions in a simple polynomial function, which are easier to manipulate

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + o\Big((x - x_0)^n\Big) \quad \text{as } x \to x_0$$

But we will mostly limit our study cases at $x \to 0$ (for now)

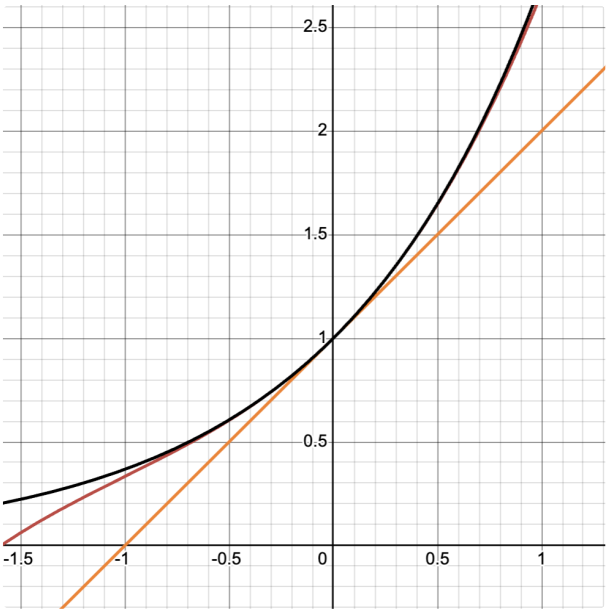
$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^k + o(x^n) \quad \text{as } x \to 0$$

For now, since derivatives are not yet introduced, all the expansions will be provided.

Consider $f(x) = e^x$ at $x_0 = 0$. Its expansion is (up to order 3) $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$.
Now let's consider how well the various orders approximate $e^{0.5} = 1.64872$

Order n	$P_n(x)$	$P_n(0.5)$	$e^{0.5} - P_n(0.5)$
0	1	1	0.649
1	$1 + x$	1.5	0.149
2	$1 + x + \frac{1}{2}x^2$	1.625	0.024
3	$1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$	1.64583	0.0029

at order 3 it is already 99.8% accurate!



f(x) in black, 1st order expansion in orange, 3rd order expansion in red

4.2.1) Little-o Notation

Little o is the correction term. This encapsulates the difference of $f(x)$ and $P_n(x)$.

Definition 1 (Little-o)**.**

Given two functions $f(x)$ and $g(x)$ near x_0 (with $g(x) \neq 0$ in a punctured neighborhood of x_0), we say that $f(x)$ **is "little-o" of** $g(x)$ **as** $x \to x_0$, written $f(x) = o\Big(g(x)\Big)$, if:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0$$

This is, in the Taylor expansion writing $o(x^n)$ means that all the next order terms go to zero faster than x^n and can be ignored.

For now let's just focus on $x \to 0$

Intuitively, little-o, refers to any function that shrinks faster then the specified function. Therefore we can say that $f(x) = o(g(x))$ is an abuse of notation as it should be $f(x) \in o(g(x))$ since $o(g(x))$ is a set that refers to all functions $h(x)$ where $h(x) \ll g(x)$

For example consider powers. As $x \to 0$ we have

$$x^n \ll x^m \quad \text{for } n > m$$

Therefore we can say:

- $x^3 = o(x^2)$ since $\lim_{x \to 0} x^3/x^2 = \lim_{x \to 0} x = 0$
- But not $x^2 = o(x^3)$ since $\lim_{x \to 0} x^2/x^3 = \lim_{x \to 0} 1/x = \pm\infty$

Theorem 2 (Little-o Properties)**.**

Here are some properties of little-o. Let $n > 0$*:*

- *Constant Multiplication:*

$$c \cdot o(x^n) = o(x^n) \text{ for any constant } c \neq 0 \qquad \text{The error has the same order}$$

- *Same Order Addition/subtraction:*

$$o(x^n) \pm o(x^n) = o(x^n) \quad \text{Important: } o(x^n) - o(x^n) \text{ is NOT } =0$$

- *Different order addition (Absorption):*

$$o(x^n) + o(x^m) = o(x^{\min(n,m)}) \quad \text{Ex: } o(x^2) + o(x^4) = o(x^2) \text{ since } x^4 \in o(x^2)$$

- *Absorbing Powers:*

$$c \cdot x^m + o(x^n) = o(x^n) \text{ if } m > n \quad \text{Ex: } 5x^7 + o(x^3) = o(x^3) \text{ since } x^7 \in o(x^3)$$

- *Multiplication by powers:*

$$x^m \cdot o(x^n) = o(x^{m+n}) \quad \text{Ex: } x^5 \cdot o(x^3) = o(x^8)$$

- *Product of two Little-o's:*

$$o(x^m) \cdot o(x^n) = o(x^{m+n}) \quad \text{Ex: } o(x^2) \cdot o(x^3) = o(x^5)$$

- *Powers of Little-o:*

$$[o(x^n)]^k = o(x^{n \cdot k}) \quad \text{Ex: } [o(x^2)]^3 = o(x^6)$$

Proofs (very quick):

1. Constant Multiplication: $c \cdot o(x^n) = o(x^n)$ $(c \neq 0)$
   Let $f(x) \in o(x^n)$, consider $g(x) = c \cdot f(x)$:

$$\lim_{x \to 0} \frac{g(x)}{x^n} = \lim_{x \to 0} \frac{c \cdot f(x)}{x^n} = c \cdot \lim_{x \to 0} \frac{f(x)}{x^n} = c \cdot 0 = 0$$

   Therefore, $c \cdot f(x) \in o(x^n)$.
2. Same-Order Addition / Subtraction: $o(x^n) \pm o(x^n) = o(x^n)$
   Let $f(x), g(x) \in o(x^n)$, consider $h(x) = f(x) \pm g(x)$:

$$\lim_{x \to 0} \frac{h(x)}{x^n} = \lim_{x \to 0} \left(\frac{f(x)}{x^n} \pm \frac{g(x)}{x^n} \right) = \lim_{x \to 0} \frac{f(x)}{x^n} \pm \lim_{x \to 0} \frac{g(x)}{x^n} = 0 \pm 0 = 0$$

   Hence, $f(x) \pm g(x) \in o(x^n)$.
3. Different-Order Addition (Absorption): $o(x^n) + o(x^m) = o(x^{\min(n,m)})$
   Assume (without loss of generality) that $n \leq m$, so $\min(n, m) = n$.
   Let $f(x) \in o(x^n)$ and $g(x) \in o(x^m)$.
   Examine the quotient with x^n:

$$\lim_{x \to 0} \frac{f(x) + g(x)}{x^n} = \lim_{x \to 0} \frac{f(x)}{x^n} + \lim_{x \to 0} \frac{g(x)}{x^n}$$

   By assumption, $\lim_{x \to 0} \frac{f(x)}{x^n} = 0$.
   For $g(x)$, rewrite the fraction by factoring out x^{m-n} (where $m - n \geq 0$):

$$\lim_{x \to 0} \frac{g(x)}{x^n} = \lim_{x \to 0} \left(\frac{g(x)}{x^m} \cdot x^{m-n} \right) = \left(\lim_{x \to 0} \frac{g(x)}{x^m} \right) \cdot \left(\lim_{x \to 0} x^{m-n} \right) = 0 \cdot \lim_{x \to 0} x^{m-n} = 0$$

   Thus:

$$\lim_{x\to 0}\frac{f(x)+g(x)}{x^n}=0+0=0 \implies f(x)+g(x)\in o(x^n)$$

4. Absorbing Powers: $c\cdot x^m+o(x^n)=o(x^n)$ if $m>n$
   Let $f(x)\in o(x^n)$ and $h(x)=cx^m+f(x)$ with $m>n$.
   Compute the limit:

$$\lim_{x\to 0}\frac{h(x)}{x^n}=\lim_{x\to 0}\left(\frac{cx^m}{x^n}+\frac{f(x)}{x^n}\right)=\lim_{x\to 0}\left(cx^{m-n}+\frac{f(x)}{x^n}\right)$$

   Because $m-n>0$:

$$\lim_{x\to 0}cx^{m-n}=0 \quad \text{and} \quad \lim_{x\to 0}\frac{f(x)}{x^n}=0$$

   Adding the limits yields $0+0=0$, confirming $cx^m+f(x)\in o(x^n)$.
5. Multiplication by Powers: $x^m\cdot o(x^n)=o(x^{m+n})$
   Let $f(x)\in o(x^n)$. Consider $g(x)=x^m\cdot f(x)$:

$$\lim_{x\to 0}\frac{g(x)}{x^{m+n}}=\lim_{x\to 0}\frac{x^m\cdot f(x)}{x^m\cdot x^n}=\lim_{x\to 0}\frac{f(x)}{x^n}=0$$

   Hence, $x^m\cdot f(x)\in o(x^{m+n})$.
6. Product of Two Little-o's: $o(x^m)\cdot o(x^n)=o(x^{m+n})$
   Let $f(x)\in o(x^m)$ and $g(x)\in o(x^n)$. Consider $h(x)=f(x)\cdot g(x)$:

$$\lim_{x\to 0}\frac{h(x)}{x^{m+n}}=\lim_{x\to 0}\left(\frac{f(x)}{x^m}\cdot\frac{g(x)}{x^n}\right)=\left(\lim_{x\to 0}\frac{f(x)}{x^m}\right)\cdot\left(\lim_{x\to 0}\frac{g(x)}{x^n}\right)=0\cdot 0=0$$

   Hence, $f(x)\cdot g(x)\in o(x^{m+n})$.
7. Powers of Little-o: $[o(x^n)]^k=o(x^{n\cdot k})$ ($k\in\mathbb{N}_{\geq 1}$)
   Let $f(x)\in o(x^n)$. Consider $g(x)=[f(x)]^k$:

$$\lim_{x\to 0}\frac{g(x)}{x^{n\cdot k}}=\lim_{x\to 0}\frac{[f(x)]^k}{(x^n)^k}=\lim_{x\to 0}\left(\frac{f(x)}{x^n}\right)^k$$

   By the continuous mapping theorem (or the product rule applied k times):

$$\lim_{x\to 0}\left(\frac{f(x)}{x^n}\right)^k=\left(\lim_{x\to 0}\frac{f(x)}{x^n}\right)^k=0^k=0$$

   Thus, $[f(x)]^k\in o(x^{n\cdot k})$.

4.3) Exercises

1) Little-*o* Warm Up Exercise:

Simplify the following expression as $x\to 0$:

$$E(x)=\left(x+2x^2+o(x^2)\right)-\left(x-x^2+4x^3+o(x^3)\right)$$

Do the basic algebraic simplifications:

$$E(x)=3x^2-4x^3+o(x^2)-o(x^3)$$

Since $o(x^2)+o(x^3)=o(x^2)$ we have

$$E(x)=3x^2-4x^3+o(x^2)$$

But notice that now $4x^3\in o(x^2)$ and thus

$$E(x)=3x^2+o(x^2)$$

2) Classic Cancellation Limit

Compute the limit:

$$\lim_{x\to 0}\frac{\sin(x)-x}{x^3}$$

Recall the expansion for sin:

$$\sin(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!}=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots$$

This cannot be computed via the notable limit since:

$$\lim_{x\to 0}\frac{\sin(x)-x}{x^3}=\lim_{x\to 0}\frac{\sin(x)}{x^3}-\frac{1}{x^2}=\lim_{x\to 0}\frac{\sin(x)}{x}\cdot\frac{1}{x^2}-\frac{1}{x^2}$$

Where it is known that $\lim_{x\to 0}\sin(x)/x=1$ therefore:

$$\lim_{x\to 0}\frac{\sin(x)}{x}\cdot\frac{1}{x^2}-\frac{1}{x^2}=\lim_{x\to 0}\frac{1}{x^2}-\frac{1}{x^2}=0$$

But the last step is illegal, as the limit cannot be evaluated partially (that is we cannot simplify out $\sin(x)/x$ but keep $1/x^2$)

Try with order 1 expansion:

$$\lim_{x\to 0}\frac{\sin(x)-x}{x^3}=\lim_{x\to 0}\frac{x+o(x)-x}{x^3}=\lim_{x\to 0}\frac{o(x)}{x^3}$$

but since $o(x)/x^3=x^{-3}o(x)=o(x^{-2})$ we cannot determine it.

Since the denominator is of order 3, we expand until this order:

$$\lim_{x\to 0}\frac{x-\frac{x^3}{6}+o(x^3)-x}{x^3}=\lim_{x\to 0}\frac{-\frac{x^3}{6}+o(x^3)}{x^3}=\lim_{x\to 0}-\frac{1}{6}+\frac{o(x^3)}{x^3}=-\frac{1}{6}+0=-\frac{1}{6}$$

since $o(x^3)/x^3=x^{-3}o(x^3)=o(1)\to 0$ as $x\to 0$ and more in general $\lim_{x\to 0}o(x^n)/x^m=0$ if $n\geq m$.

3) Two Expansions:

Compute the limit:

$$\lim_{x\to 0}\frac{e^x-\cos(x)-x}{x^2}$$

Recall both expansions:

$$e^x=\sum_{n=0}^{\infty}\frac{x^n}{n!}=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots$$
$$\cos(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$$

Apply them until order 2

$$\begin{aligned}e^x-\cos(x)-x&=\left(1+x+\frac{x^2}{2}+o(x^2)\right)-\left(1-\frac{x^2}{2}+o(x^2)\right)-x\\&=1-1+x-x+\frac{x^2}{2}+\frac{x^2}{2}+o(x^2)+o(x^2)\\&=x^2+o(x^2)\end{aligned}$$

Then the limit becomes:

$$\lim_{x\to 0}\frac{e^x-\cos(x)-x}{x^2}=\lim_{x\to 0}\frac{x^2+o(x^2)}{x^2}=1+0=1$$

4) Substitution

Find the Taylor expansion of order 4 at $x_0=0$ for $g(x)=e^{-x^2}$ and use it to compute:

$$\lim_{x\to 0}\frac{e^{-x^2}-\cos(x)}{x^4}$$

Recall both expansions:

$$e^x=\sum_{n=0}^{\infty}\frac{x^n}{n!}=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots$$
$$\cos(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$$

We need to expand up to order 4. For cosine it is known, but e^{-x^2} doesn't have the basic formula.
Notice that with $t=-x^2$ we can expand e^t up to order 2 (since $o(t^2)\to o(x^4)$). This can be done since both tend to 0 as $x\to 0$

$$e^t=1+t+\frac{t^2}{2}+o(t^2)\to e^{-x^2}=1-x^2+\frac{x^4}{2}+o(x^4)$$

Now expand the limit:

$$\begin{aligned}\lim_{x\to 0}\frac{e^{-x^2}-\cos(x)}{x^4}&=\lim_{x\to 0}\frac{1-x^2+\frac{x^4}{2}+o(x^4)-1+\frac{x^2}{2}-\frac{x^4}{24}-o(x^4)}{x^4}\\&=\lim_{x\to 0}\frac{-\frac{1}{2}x^2+\frac{11}{24}x^4+o(x^4)}{x^4}\\&=\lim_{x\to 0}\frac{11}{24}-\frac{1}{2x^2}+\frac{o(x^4)}{x^4}\\&=\frac{11}{24}-\infty+0\\&=-\infty\end{aligned}$$

5) Exam 14.09.2020

Compute the limit:

$$\lim_{x\to 0^+}\frac{\log(1+\sinh x)-\sin x}{x^2}$$

Recall the expansions:

$$\sinh(x)=x+\frac{x^3}{6}+o(x^3)$$
$$\log(1+t)=t-\frac{t^2}{2}+o(t^2)$$
$$\sin(x)=x-\frac{x^3}{6}+o(x^3)$$

Here expansion of order two is needed. For the sin it's straight forward.
The log of sinh is done by substituting $t=\sinh(x)$, expanding the log and then the $\sinh(x)$.
How much do we need to expand?

$$\begin{aligned}\log(1+t) = t + o(t) &= \sinh(x) + o(\sinh(x)) \\ &= x + \frac{x^3}{6} + o(x^3) + o\left(x + \frac{x^3}{6} + o(x^3)\right) \\ &= x + \frac{x^3}{6} + o(x) \\ &= x + o(x)\end{aligned}$$

So not enough, in general if order n is needed, the expansion must be of order n

$$\begin{aligned}\log(1+t) = t - \frac{t^2}{2} + o(t^2) &= \sinh(x) - \frac{\sinh^2(x)}{2} + o(\sinh^2(x)) \\ &= x + o(x^2) - \frac{(x+o(x^2))^2}{2} + o([x+o(x^2)]^2) \\ &= x + o(x^2) - \frac{x^2 + 2xo(x^2) + o(x^2)^2}{2} + o(x^2 + o(x^3) + o(x^4)) \\ &= x - \frac{x^2}{2} + o(x^2)\end{aligned}$$

Now expand the whole limit:

$$\begin{aligned}\lim_{x\to 0^+} \frac{\log(1+\sinh x) - \sin x}{x^2} &= \lim_{x\to 0^+} \frac{x - \frac{x^2}{2} + o(x^2) - x + o(x^2)}{x^2} \\ &= \lim_{x\to 0^+} \frac{-\frac{x^2}{2} + o(x^2)}{x^2} \\ &= \lim_{x\to 0^+} -\frac{1}{2} + \frac{o(x^2)}{x^2} = -\frac{1}{2}\end{aligned}$$

5) Lesson 27/10/26

Warmup Exercise

Compute the limit:

$$\lim_{x\to 0} \frac{\log(\cos(x)) + \frac{x^2}{2}}{x^4}$$

Recall the expansions:

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$
$$\log(1+t) = t - \frac{t^2}{2} + o(t^2)$$

Since the denominator is of order 4, we must expand the numerator up to $o(x^4)$.

Notice that $\log(\cos(x))$ is not a standard expansion, as the known formula is for $\log(1+t)$ as $t \to 0$. We cannot simply substitute $t = \cos(x)$ because $\cos(x) \to 1 \neq 0$ as $x \to 0$.

To fix this, we add and subtract 1 inside the logarithm, rewriting it as $\log\Big(1 + (\cos(x) - 1)\Big)$. Now the natural substitution is $t = \cos(x) - 1$, which is valid since $t \to 0$ as $x \to 0$.

Expanding $\cos(x)$ up to order 4 gives:

$$t = \cos(x) - 1 = -\frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

Notice that the lowest degree in t is x^2. Therefore, $\log(1+t)$ only needs to be expanded up to degree 2 in t (since $o(t^2) = o\Big((x^2)^2\Big) = o(x^4)$). Substituting t into $\log(1+t) = t - \frac{t^2}{2} + o(t^2)$, we get:

$$t = \cos(x) - 1 \quad \to \quad t = -\frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

Now notice that the lowest degree in t is x^2, so the $\log(1+t)$ just needs to be expanded to the second degree (since $o(x^2)^2 = o(x^4)$) and compute:

$$\begin{aligned}
\log(\cos(x)) = t - \frac{t^2}{2} + o(t^2) &= \left(-\frac{x^2}{2} + \frac{x^4}{24} + o(x^4)\right) - \frac{1}{2}\left(-\frac{x^2}{2} + \frac{x^4}{24} + o(x^4)\right)^2 + o(x^4) \\
&= -\frac{x^2}{2} + \frac{x^4}{24} - \frac{1}{2}\left(\frac{x^4}{4} + o(x^4)\right) + o(x^4) \\
&= -\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^4}{8} + o(x^4) \\
&= -\frac{x^2}{2} - \frac{x^4}{12} + o(x^4)
\end{aligned}$$

Now replace the expansion into the whole limit:

$$\begin{aligned}
\lim_{x\to 0} \frac{\log(\cos(x)) + \frac{x^2}{2}}{x^4} &= \lim_{x\to 0} \frac{-\frac{x^2}{2} - \frac{x^4}{12} + o(x^4) + \frac{x^2}{2}}{x^4} \\
&= \lim_{x\to 0} \frac{-\frac{x^4}{12} + o(x^4)}{x^4} \\
&= \lim_{x\to 0} \left(-\frac{1}{12} + \frac{o(x^4)}{x^4}\right) \\
&= -\frac{1}{12} + 0 = -\frac{1}{12}
\end{aligned}$$

6) Exam 17.09.2019

Compute the limit:

$$\lim_{x\to 0^+} \frac{e^{x-2x^2}-1-x}{\sinh(x^2)+x^{7/3}\log x}$$

Recall the expansions:

$$\sinh(t) = t + \frac{t^3}{6} + o(t^3)$$
$$e^t = 1 + t + \frac{t^2}{2} + o(t^2)$$

Denominator:
First, look at $\sinh(x^2)$. Using $t = x^2$ (since $t \to 0$ as $x \to 0^+$), its leading term is of order 2:

$$\sinh(x^2) = \sinh(t) = t + o(t^2) = x^2 + o(x^4) = x^2 + o(x^2)$$

Next, look at $x^{7/3}\log(x)$. Since $\log(x)$ has no Taylor expansion at 0, we check whether this term is negligible compared to the leading x^2 term, that is, whether $x^{7/3}\log(x) \in o(x^2)$. Recall that $\lim_{x\to 0^+} x^\alpha \log x = 0$ for any $\alpha > 0$:

$$\lim_{x\to 0^+} \frac{x^{7/3}\log(x)}{x^2} = \lim_{x\to 0^+} x^{1/3}\log(x) = 0 \implies x^{7/3}\log(x) = o(x^2)$$

Combining both terms in the denominator and applying the absorption rule

$$\sinh(x^2) + x^{7/3}\log x = x^2 + o(x^4) + o(x^2) = x^2 + o(x^2)$$

Numerator:
Since the denominator is of order 2 and the $1 + x$ terms cancel out in the numerator, we expand e^t with $t = x - 2x^2$ up to order 2. Notice that $o(t^2) = o([x-2x^2]^2) = o(x^2)$, which absorbs all powers x^3 and higher:

$$\begin{aligned} e^{x-2x^2} = 1 + t + \frac{t^2}{2} + o(t^2) &= 1 + (x-2x^2) + \frac{(x-2x^2)^2}{2} + o([x-2x^2]^2) \\ &= 1 + x - 2x^2 + \frac{x^2 - 4x^3 + 4x^4}{2} + o(x^2) \\ &= 1 + x - 2x^2 + \frac{1}{2}x^2 + o(x^2) \\ &= 1 + x - \frac{3}{2}x^2 + o(x^2) \end{aligned}$$

Whole limit:

$$\begin{aligned} \lim_{x\to 0^+} \frac{e^{x-2x^2}-1-x}{\sinh(x^2)+x^{7/3}\log x} &= \lim_{x\to 0^+} \frac{1 + x - \frac{3}{2}x^2 + o(x^2) - 1 - x}{x^2 + o(x^2)} \\ &= \lim_{x\to 0^+} \frac{-\frac{3}{2}x^2 + o(x^2)}{x^2 + o(x^2)} \\ &= \lim_{x\to 0^+} \frac{x^2\left(-\frac{3}{2} + \frac{o(x^2)}{x^2}\right)}{x^2\left(1 + \frac{o(x^2)}{x^2}\right)} \\ &= \frac{-\frac{3}{2} + 0}{1 + 0} = -\frac{3}{2} \end{aligned}$$

7) Exam

Compute the limit:

$$\lim_{x\to 0}\frac{1-\cos(x^2)}{x(x-\sin x)+x^3\sin^2 x}$$

Recall the expansions:

$$\cos(t)=1-\frac{t^2}{2}+o(t^2)$$
$$\sin(x)=x-\frac{x^3}{6}+o(x^4)$$

Denominator:
If we only expand $\sin(x)$ to order 1, the first term in the denominator cancels out completely and leaves us with no leading term:

$$x(x-\sin x)=x\Big(x-(x+o(x))\Big)=x\cdot o(x)=o(x^2)$$

So we must expand $\sin(x)$ to the next non-zero term (order 3):

$$x(x-\sin x)=x\left(x-x+\frac{x^3}{6}+o(x^3)\right)=\frac{x^4}{6}+x\cdot o(x^3)=\frac{x^4}{6}+o(x^4)=\frac{x^4}{6}+o(x^4)$$

Now check the second term, $x^3\sin^2(x)$. We can see it is $o(x^4)$ either via the definition (using notable limit):

$$\lim_{x\to 0}\frac{x^3\sin^2(x)}{x^4}=\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^2\cdot x=1^2\cdot 0=0\implies x^3\sin^2(x)=o(x^4)$$

or directly via little-o algebra

$$\sin x=x+o(x)\implies x^3\sin^2 x=x^3(x+o(x))^2=x^5+o(x^5)=o(x^4)$$

Therefore, the denominator becomes:

$$x(x-\sin x)+x^3\sin^2 x=\frac{x^4}{6}+o(x^4)$$

Numerator:
Using $t=x^2$ in the cosine expansion up to order 2 in t (which gives order 4 in x):

$$1-\cos(x^2)=1-\left(1-\frac{(x^2)^2}{2}+o((x^2)^2)\right)=\frac{x^4}{2}+o(x^4)$$

Whole limit:

$$\begin{aligned}\lim_{x\to 0}\frac{1-\cos(x^2)}{x(x-\sin x)+x^3\sin^2 x}&=\lim_{x\to 0}\frac{\frac{x^4}{2}+o(x^4)}{\frac{x^4}{6}+o(x^4)}\\&=\lim_{x\to 0}\frac{x^4\left(\frac{1}{2}+\frac{o(x^4)}{x^4}\right)}{x^4\left(\frac{1}{6}+\frac{o(x^4)}{x^4}\right)}\\&=\frac{1/2}{1/6}=3\end{aligned}$$

8) Exam 12.09.2022

Compute the limit $\forall\alpha\in\mathbb{R}$:

$$\lim_{x\to 0^+} \frac{\sin x - \alpha x + \frac{1}{6}\alpha x^3}{\arctan(x^2+4x^3)}$$

First recall the standard expansions:

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5)$$
$$\arctan(t) = t - \frac{t^3}{3} + \frac{t^5}{5} + o(t^5)$$

Numerator:
Since the polynomial terms go up to x^3, we expand $\sin(x)$ up to order 3:

$$\sin x - \alpha x + \frac{\alpha}{6}x^3 = x - \frac{x^3}{6} + o(x^3) - \alpha x + \frac{\alpha}{6}x^3 = (1-\alpha)x + \frac{\alpha - 1}{6}x^3 + o(x^3)$$

Denominator:
Use the substitution $t = x^2 + 4x^3$ (since $t \to 0$ as $x \to 0^+$) and expand $\arctan(t)$ to order 1.

$$\arctan(x^2+4x^3) = t + o(t) = x^2 + 4x^3 + o(x^2) = x^2 + o(x^2)$$

Now rewrite the entire limit:

$$\lim_{x\to 0^+} \frac{\sin x - \alpha x + \frac{1}{6}\alpha x^3}{\arctan(x^2+4x^3)} = \lim_{x\to 0^+} \frac{(1-\alpha)x + \frac{\alpha-1}{6}x^3 + o(x^3)}{x^2 + o(x^2)}$$

Now we split by cases depending on whether the leading terms in the numerator cancel out:

- **Case** $\alpha = 1$ (both the x and x^3 terms vanish):

$$\begin{aligned}\lim_{x\to 0^+} \frac{o(x^3)}{x^2 + o(x^2)} &= \lim_{x\to 0^+} \frac{o(x^3)}{x^2\left(1 + \frac{o(x^2)}{x^2}\right)} \\ &= \lim_{x\to 0^+} \frac{o(x^3)}{x^2} \cdot \frac{1}{1 + \frac{o(x^2)}{x^2}} \\ &= 0 \cdot \frac{1}{1+0} = 0\end{aligned}$$

  (since $\frac{o(x^3)}{x^2} = o(x) \to 0$ as $x \to 0^+$)
- **Case** $\alpha \neq 1$ (the order-1 term $(1-\alpha)x$ dominates the numerator):
  Factor out x from numerator and denominator:

$$\lim_{x\to 0^+} \frac{x\left[(1-\alpha) + \frac{\alpha-1}{6}x^2 + \frac{o(x^3)}{x}\right]}{x\left[x + \frac{o(x^2)}{x}\right]} = \lim_{x\to 0^+} \frac{1 - \alpha + \frac{\alpha-1}{6}x^2 + o(x^2)}{x + o(x)}$$
$$= \frac{1-\alpha}{0^+} = \begin{cases} +\infty & \text{if } \alpha < 1 \\ -\infty & \text{if } \alpha > 1 \end{cases}$$

So the final result is:

$$\lim_{x\to 0^+} \frac{\sin(x) - \alpha x + \frac{1}{6}\alpha x^3}{\arctan(x^2+4x^3)} = \begin{cases} +\infty & \text{if } \alpha < 1 \\ 0 & \text{if } \alpha = 1 \\ -\infty & \text{if } \alpha > 1 \end{cases}$$

In class however (see written notes) we aimed to have same order at numerator and denominator, however the result was the same if not with a bit more algebra.

6) Lesson 6/11/26

6.1) Standard Limit Exercise

5) Exponential (Harder)

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = e^{\frac{x^2-4}{x-1}}$$

Domain:

$$D = \{x - 1 \neq 0\} = \mathbb{R} \setminus \{1\} = (-\infty, 1) \cup (1, +\infty)$$

Sign:
Since the exponential function satisfies $e^t > 0$ for all $t \in \mathbb{R}$:

$$f(x) > 0 \quad \forall x \in D$$

We don't have any zeros.

Limits at infinity:

- Positive Infinity:

  $$\lim_{x \to +\infty} f(x) = e^{+\infty} = +\infty$$

  Is there an oblique asymptote?
  Check the slope m:

  $$m = \lim_{x \to +\infty} \frac{f(x)}{x} = \lim_{x \to +\infty} \frac{e^{\frac{x^2-4}{x-1}}}{x}$$

  Notice by polynomial division that $\frac{x^2-4}{x-1} = x + 1 - \frac{3}{x-1}$, so:

  $$\frac{f(x)}{x} = \frac{e^{x+1} \cdot e^{-\frac{3}{x-1}}}{x} = e \cdot e^{-\frac{3}{x-1}} \cdot \frac{e^x}{x}$$

  Therefore

  $$\lim_{x \to \infty} \frac{f(x)}{x} = e \cdot e^0 \cdot \infty = \infty$$

  There is no horizontal or oblique asymptote as $x \to +\infty$.
- Negative Infinity:

  $$\lim_{x \to -\infty} f(x) = e^{-\infty} = 0^+$$

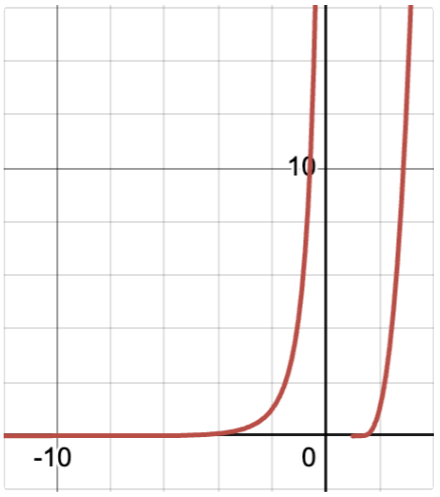
  We have a horizontal asymptote at $y = 0$ as $x \to -\infty$.

Limits at accumulation points:

- At 1

  $$\lim_{x \to 1^+} \frac{x^2 - 4}{x - 1} = \frac{-3}{0^+} = -\infty \implies \lim_{x \to 1^+} f(x) = e^{-\infty} = 0^+$$

  $$\lim_{x \to 1^-} \frac{x^2 - 4}{x - 1} = \frac{-3}{0^-} = +\infty \implies \lim_{x \to 1^-} f(x) = e^{+\infty} = +\infty$$



Graph

Polynomial division:
Add and Subtract (Difference of Squares Trick)
To make a multiple of the denominator $(x-1)$ appear in the numerator, recall that $x^2-1=(x-1)(x+1)$. We rewrite -4 as $-1-3$:

$$\begin{aligned}\frac{x^2-4}{x-1} &= \frac{(x^2-1)-3}{x-1}\\ &= \frac{x^2-1}{x-1}-\frac{3}{x-1}\\ &= \frac{(x-1)(x+1)}{x-1}-\frac{3}{x-1}\\ &= x+1-\frac{3}{x-1}\end{aligned}$$

6) Sqrt (Hardest)

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x)=5x+2-\sqrt{25x^2+12x}$$

Domain:
The roots of the argument of the sqrt are $x=-\frac{12}{25}$ and $x=0$. Taking the external intervals:

$$D=\left\{25x^2+12x\geq 0\right\}=\left(-\infty,-\frac{12}{25}\right]\cup[0,+\infty)$$

Sign:

$$f(x)>0 \iff \sqrt{25x^2+12x}<5x+2$$

As seen in first class (case $\sqrt{A(x)}<B(x)$) we have:

$$\begin{cases}A(x)\geq 0\\ B(x)>0\\ A(x)<[B(x)]^2\end{cases} \iff \begin{cases}x\in\left(-\infty,-\frac{12}{25}\right]\cup[0,+\infty)\\ 5x+2>0 \implies x>-\frac{2}{5}\\ 25x^2+12x<25x^2+20x+4 \implies x>-\frac{1}{2}\end{cases}$$

Then:

$$f(x)>0 \iff x\in[0,+\infty)$$

Limits at infinity:

- Positive Infinity:

$$\lim_{x\to+\infty} f(x) = (+\infty) - (+\infty) \quad \text{indeterminate form!}$$

Rationalize the expression:
To rationalize the idea is as follows:

$$a \pm \sqrt{b} = (a \pm \sqrt{b})\frac{a \mp \sqrt{b}}{a \mp \sqrt{b}} = \frac{a^2 - b}{a \mp \sqrt{b}}$$

Apply it directly

$$\begin{aligned} f(x) &= 2 + (\underbrace{5x}_{a} - \sqrt{\underbrace{25x^2 + 12x}_{b}}) \\ &= 2 + \frac{(5x - \sqrt{25x^2+12x})(5x + \sqrt{25x^2+12x})}{5x + \sqrt{25x^2+12x}} \\ &= 2 + \frac{25x^2 - (25x^2 + 12x)}{5x + \sqrt{25x^2+12x}} \qquad \$\$For \$x > 0\$, extract \$|x| = x\$: \$\$ \lim_{x\to+\infty} f(x) = \lim_{x\to+\infty} 2 \cdot \\ &= 2 - \frac{12x}{5x + \sqrt{25x^2+12x}} \\ &= 2 - \frac{12x}{5x + |x|\sqrt{25 + \frac{12}{x}}} \end{aligned}$$

- Negative Infinity:

$$\lim_{x\to-\infty} f(x) = 5(-\infty) + 2 - \sqrt{+\infty} = -\infty - \infty = -\infty$$

This is not indeterminate.
Is there an oblique asymptote?
Find the slope m (recall now since $x < 0 \to |x| = -x$):

$$m = \lim_{x\to-\infty} \frac{f(x)}{x} = \lim_{x\to-\infty} \left(5 + \frac{2}{x} - \frac{\sqrt{25x^2+12x}}{x}\right) = 5 - \lim_{x\to-\infty} \frac{-x\sqrt{25 + \frac{12}{x}}}{x} = 10$$

Now find the intercept q:

$$q = \lim_{x\to-\infty} [f(x) - 10x] = \lim_{x\to-\infty} [2 - 5x - \sqrt{25x^2+12x}] = 2 - \lim_{x\to-\infty} \left(5x + \sqrt{25x^2+12x}\right)$$

This is an indeterminate form $(-\infty) + (+\infty)$.
Rationalize

$$\begin{aligned} 5x + \sqrt{25x^2+12x} &= \left(5x + \sqrt{25x^2+12x}\right)\frac{5x - \sqrt{25x^2+12x}}{5x - \sqrt{25x^2+12x}} \\ &= \frac{25x^2 - \left(25x^2+12x\right)}{5x - \sqrt{25x^2+12x}} \\ &= \frac{-12x}{5x - \sqrt{25x^2+12x}} \\ &= \frac{-12x}{5x - |x|\sqrt{25 + \frac{12}{x}}} \end{aligned}$$

Then

$$q = 2 + \lim_{x\to-\infty} \frac{12x}{5x - |x|\sqrt{25 + \frac{12}{x}}} = 2 + \lim_{x\to-\infty} \frac{12}{5 + \sqrt{25 + \frac{12}{x}}} = 2 + \frac{12}{10} = \frac{16}{5}$$

We have an oblique asymptote:

$$y = 10x + \frac{16}{5} \quad \text{as } x \to -\infty$$

Limits at accumulation points:

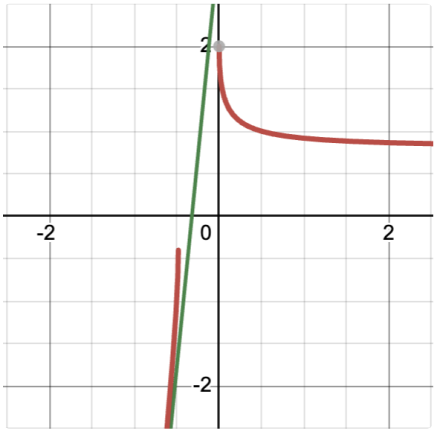
- At 0 (only 0^+ due to domain):

$$\lim_{x\to 0^+} f(x) = f(0) = 5(0) + 2 - 0 = 2$$

- At $-\frac{12}{25}$ (only $\left(-\frac{12}{25}\right)^-$ due to domain):

$$\lim_{x\to\left(-\frac{12}{25}\right)^-} f(x) = f\left(-\frac{12}{25}\right) = 5\left(-\frac{12}{25}\right) + 2 - 0 = -\frac{12}{5} + 2 = -\frac{2}{5}$$

There are no vertical asymptotes.



Asymptote in Green

6.2) Taylor Tips and Tricks

Substitution: To plug $t = f(x)$ into a standard Maclaurin expansion (centered at 0) as $x \to 0$, both x and t must tend to 0:

$$t = f(x) \text{ is valid iff } \lim_{x\to 0} f(x) = 0 \quad (\text{i.e., } t \to 0 \text{ as } x \to 0)$$

Lower Degree Change: Let $m, n \in \mathbb{R}$ such that $m < n$. As $x \to 0$:

$$o(x^n) \subset o(x^m)$$

Therefore, if needed, you can always absorb a higher-order remainder into a **lower** degree remainder, for example:

$$o(x^3) \to o(x^2) \quad \text{but NEVER} \quad o(x^2) \to o(x^3)$$

Polynomial and little-o inside little-o:
When you have a polynomial (or additional little-o terms) inside a little-o as $x \to 0$, the **lowest** non-zero degree always wins:

$$o\left(\sum_k a_k x^k\right) = o\left(x^{\min\{k\,:\,a_k \neq 0\}}\right)$$

For example:

$$o(x^2 + 3x^3 + o(x^4)) = o(x^2)$$

What degree? In general, the desired degree to expand to is the smallest one where the leading terms do not cancel out completely. If expanding $f(x) + g(x)$ to a low order n causes full cancellation:

$$f(x) + g(x) = o(x^n) \quad (\text{indeterminate if divided by } x^m)$$

you must expand to a higher order $m > n$ until a non-zero polynomial part $h(x) \neq 0$ survives:

$$f(x)+g(x)=h(x)+o(x^m)$$

In practice, first expand the simpler side (usually the denominator) until you find its first non-zero term $cx^n + o(x^n)$. Then aim to expand the numerator to at least that same order $o(x^n)$ so that, even if all terms in the numerator cancel out, you avoid indeterminate remainders of the form $\frac{o(x^m)}{x^n} = o(x^{m-n})$ with $m < n$.

6.3) Taylor Limit Exercises

Exercise

Compute the limit:

$$\lim_{x\to 0} \frac{(1-\cosh(x))^2}{\sin(x)(x-\arctan(x)) + x^3\sinh^2(x)}$$

Recall the expansions:

$$\begin{aligned}
\cosh(x) &= 1 + \frac{x^2}{2} + \frac{x^4}{24} + o(x^4) \\
\sin(x) &= x - \frac{x^3}{6} + o(x^3) \\
\arctan(x) &= x - \frac{x^3}{3} + o(x^3) \\
\sinh(x) &= x + \frac{x^3}{6} + o(x^3)
\end{aligned}$$

Numerator:
The numerator needs to be expanded at least to order two (otherwise the terms would cancel out completely)

$$1-\cosh(x) = 1 - \left(1 + \frac{x^2}{2} + o(x^2)\right) = -\frac{x^2}{2} + o(x^2)$$

Now square the expression:

$$(1-\cosh(x))^2 = \left(-\frac{x^2}{2} + o(x^2)\right)^2 = \frac{x^4}{4} - x^2 o(x^2) + [o(x^2)]^2 = \frac{x^4}{4} + o(x^4)$$

So the numerator is of order 4, denominator must be 4 or less!

Denominator:
Since the numerator is of order 4, we aim to expand the denominator up to $o(x^4)$.

Look at the first term, $\sin(x)(x - \arctan(x))$. Notice that expanding $\arctan(x)$ only to order 1 or 2 causes a cancellation inside the bracket:

$$x - \arctan(x) = x - (x + o(x^2)) = o(x^2)$$

So we must expand $\arctan(x)$ to order 3:

$$x - \arctan(x) = x - \left(x - \frac{x^3}{3} + o(x^3)\right) = \frac{x^3}{3} + o(x^3)$$

Since this bracket already starts at x^3, we only need to expand the outer factor $\sin(x)$ to order 1 ($\sin(x) = x + o(x)$) so that their product reaches order 4:

$$\begin{aligned}\sin(x)(x-\arctan(x)) &= (x+o(x))\left(\frac{x^3}{3}+o(x^3)\right)\\ &= \frac{x^4}{3}+x\cdot o(x^3)+\frac{x^3}{3}o(x)+o(x)o(x^3)\\ &= \frac{x^4}{3}+o(x^4)\end{aligned}$$

Now notice that the second term in the denominator, $x^3\sinh^2(x)$, is also $o(x^4)$ since:

$$\lim_{x\to 0}\frac{x^3\sinh^2(x)}{x^4}=\lim_{x\to 0}\left(\frac{\sinh(x)}{x}\right)^2\cdot x = 1^2\cdot 0 = 0 \implies x^3\sinh^2(x)=o(x^4)$$

(or directly via little-o algebra: $x^3(x+o(x))^2 = x^5+o(x^5)=o(x^4)$).

Therefore, the denominator becomes:

$$\sin(x)(x-\arctan(x))+x^3\sinh^2(x)=\frac{x^4}{3}+o(x^4)+o(x^4)=\frac{x^4}{3}+o(x^4)$$

Whole limit:

$$\begin{aligned}\lim_{x\to 0}\frac{(1-\cosh(x))^2}{\sin(x)(x-\arctan(x))+x^3\sinh^2(x)} &= \lim_{x\to 0}\frac{\frac{x^4}{4}+o(x^4)}{\frac{x^4}{3}+o(x^4)}\\ &= \lim_{x\to 0}\frac{x^4\left(\frac{1}{4}+\frac{o(x^4)}{x^4}\right)}{x^4\left(\frac{1}{3}+\frac{o(x^4)}{x^4}\right)}\\ &= \frac{1/4}{1/3}=\frac{3}{4}\end{aligned}$$

Parametric

Compute the limit $\forall\alpha>0$:

$$\lim_{x\to 0^+}\frac{\arctan(x)-\sin(x)+x^{10/3}\log(x)}{x^\alpha(1-\cos^2(x))}$$

First recall the standard expansions (as $x\to 0$) and standard logarithmic limit:

$$\begin{aligned}\sin(x) &= x-\frac{x^3}{6}+\frac{x^5}{120}+o(x^5)\\ \arctan(x) &= x-\frac{x^3}{3}+\frac{x^5}{5}+o(x^5)\\ \lim_{x\to 0^+}x^\beta\log(x) &= 0 \qquad \forall\beta>0\end{aligned}$$

Numerator:
Since the order-1 terms of $\arctan(x)$ and $\sin(x)$ cancel out ($x-x=0$), we must expand both functions up to order 3 (the first non-zero term):

$$\arctan(x)-\sin(x)=\left(x-\frac{x^3}{3}+o(x^3)\right)-\left(x-\frac{x^3}{6}+o(x^3)\right)=-\frac{1}{6}x^3+o(x^3)$$

Now let's compare the remaining term $x^{10/3}\log(x)$ with x^3. Since $\frac{10}{3}>3$ and polynomial decay dominates logarithmic growth as $x\to 0^+$, we can verify that $x^{10/3}\log(x)=o(x^3)$ by checking the definition of little-o:

$$\lim_{x\to 0^+}\frac{x^{10/3}\log(x)}{x^3}=\lim_{x\to 0^+}x^{1/3}\log(x)=0 \quad\implies\quad x^{10/3}\log(x)=o(x^3)$$

Putting the numerator together:

$$\arctan(x) - \sin(x) + x^{10/3}\log(x) = -\frac{1}{6}x^3 + o(x^3) + o(x^3) = -\frac{1}{6}x^3 + o(x^3)$$

Denominator:
Using the trigonometric identity $1 - \cos^2(x) = \sin^2(x)$ and expanding $\sin(x) = x + o(x)$ to order 1:

$$x^\alpha\Big(1 - \cos^2(x)\Big) = x^\alpha \sin^2(x) = x^\alpha\Big(x + o(x)\Big)^2 = x^\alpha\Big(x^2 + o(x^2)\Big) = x^{\alpha+2} + o(x^{\alpha+2})$$

(Note: You get the exact same result by expanding $\cos(x) = 1 - \frac{x^2}{2} + o(x^2)$ *and squaring it.)*

Now rewrite the entire limit:

$$\lim_{x\to 0^+} \frac{\arctan(x) - \sin(x) + x^{10/3}\log(x)}{x^\alpha(1 - \cos^2(x))} = \lim_{x\to 0^+} \frac{-\frac{1}{6}x^3 + o(x^3)}{x^{\alpha+2} + o(x^{\alpha+2})}$$

Factoring out the leading powers x^3 in the numerator and $x^{\alpha+2}$ in the denominator gives:

$$\lim_{x\to 0^+} \frac{x^3\left(-\frac{1}{6} + \frac{o(x^3)}{x^3}\right)}{x^{\alpha+2}\left(1 + \frac{o(x^{\alpha+2})}{x^{\alpha+2}}\right)} = \lim_{x\to 0^+} x^{1-\alpha} \cdot \frac{-\frac{1}{6} + \frac{o(x^3)}{x^3}}{1 + \frac{o(x^{\alpha+2})}{x^{\alpha+2}}}$$

Now we split by cases depending on the comparison between the numerator's order (3) and the denominator's order $(\alpha + 2)$:

- **Case** $\alpha = 1$ (numerator and denominator have the same order, $\alpha + 2 = 3$):
$$\lim_{x\to 0^+} x^0 \cdot \frac{-\frac{1}{6} + \frac{o(x^3)}{x^3}}{1 + \frac{o(x^3)}{x^3}} = \frac{-\frac{1}{6} + 0}{1 + 0} = -\frac{1}{6}$$
- **Case** $0 < \alpha < 1$ (the denominator has lower order $\alpha + 2 < 3$, so $1 - \alpha > 0$):
$$\lim_{x\to 0^+} x^{1-\alpha} \cdot \frac{-\frac{1}{6} + \frac{o(x^3)}{x^3}}{1 + \frac{o(x^{\alpha+2})}{x^{\alpha+2}}} = 0^+ \cdot \left(-\frac{1}{6}\right) = 0$$
- **Case** $\alpha > 1$ (the denominator has higher order $\alpha + 2 > 3$, so $\alpha - 1 > 0$):
$$\lim_{x\to 0^+} \frac{1}{x^{\alpha-1}} \cdot \frac{-\frac{1}{6} + \frac{o(x^3)}{x^3}}{1 + \frac{o(x^{\alpha+2})}{x^{\alpha+2}}} = \frac{-\frac{1}{6}}{0^+} = -\infty$$

So the final result is:

$$\lim_{x\to 0^+} \frac{\arctan(x) - \sin(x) + x^{10/3}\log(x)}{x^\alpha(1 - \cos^2(x))} = \begin{cases} 0 & \text{if } 0 < \alpha < 1 \\ -\frac{1}{6} & \text{if } \alpha = 1 \\ -\infty & \text{if } \alpha > 1 \end{cases}$$

Parametric (again)

Compute the limit $\forall \alpha \in \mathbb{R}$:

$$\lim_{x\to 0^+} \frac{\cosh(\alpha x) - e^{x^2} + x\log(\cos(x))}{x - \sin(x) + e^{-1/x^2}}$$

Recall the standard expansions:

$$\cosh(t) = 1 + \frac{t^2}{2} + \frac{t^4}{24} + o(t^4)$$
$$e^t = 1 + t + \frac{t^2}{2} + o(t^2)$$
$$\cos(x) = 1 - \frac{x^2}{2} + o(x^3)$$
$$\log(1+t) = t - \frac{t^2}{2} + o(t^2)$$
$$\sin(x) = x - \frac{x^3}{6} + o(x^3)$$

Denominator:
First expand $x - \sin(x)$. Since order 1 cancels out ($x - (x + o(x)) = o(x)$), we expand $\sin(x)$ up to order 3:

$$x - \sin(x) = x - \left(x - \frac{x^3}{6} + o(x^3)\right) = \frac{x^3}{6} + o(x^3)$$

Now notice that e^{-1/x^2} does not have a standard taylor expansion at 0 and the substitution technique fails since setting $t = 1/x^2$ we have that $x \to 0$ but $t \to \infty$. Again the definition must be used. For sake of exercises, instead of testing $o(x^3)$ we test the parametric limit $(o(x^n),\ n > 0)$:
Rewrite the fraction as a single exponential using $x^n = e^{n\log(x)}$ (for $x \to 0^+$):

$$\lim_{x\to 0^+} \frac{e^{-1/x^2}}{x^n} = \lim_{x\to 0^+} \frac{e^{-1/x^2}}{e^{n\log(x)}} = \lim_{x\to 0^+} e^{-\frac{1}{x^2} - n\log(x)}$$

Factor out $-\frac{1}{x^2}$ in the exponent:

$$\lim_{x\to 0^+} e^{-\frac{1}{x^2}\left(1 + nx^2\log(x)\right)} = e^{-\infty\cdot(1+0)} = e^{-\infty} = 0$$

(since $\lim_{x\to 0^+} x^2\log(x) = 0$ *by the standard hierarchy of infinities).*

Therefore we discovered that $e^{-1/x^2} = o(x^n),\ n > 0$ and we can freely chose $n = 3$ to comply with the previous expansion.
The denominator becomes therefore

$$\frac{x^3}{6} + o(x^3)$$

Numerator:
Since the denominator is of order 3, we aim to expand the numerator up to $o(x^3)$ (or stop earlier if a lower power survives):

1. For $\cosh(\alpha x)$, use $t = \alpha x$:

$$\cosh(t) = 1 + \frac{t^2}{2} + o(t^3) = 1 + \frac{(\alpha x)^2}{2} + o(\alpha^3 x^3) = 1 + \frac{\alpha^2}{2}x^2 + o(x^3)$$

   (since cosh *only has even powers, its order-3 term is* 0*).*
2. For e^{x^2}, use $t = x^2$ up to order 1 in t (since the next term is $\frac{t^2}{2} = \frac{x^4}{2} \in o(x^3)$):

$$e^t = 1 + t + \frac{t^2}{2} + o(t^2) = 1 + x^2 + \frac{x^4}{2} + o(x^4) = 1 + x^2 + o(x^3)$$

3. For $x\log(\cos(x))$, notice there is already an x in front, so we only need $\log(\cos(x))$ up to $o(x^2)$. Rewrite $\log(\cos(x)) = \log\left(1 + (\cos(x) - 1)\right)$ and use $t = \cos(x) - 1 = -\frac{x^2}{2} + o(x^2)$:

$$x\log(\cos(x)) = x\Big(t + o(t)\Big) = x\left(-\frac{x^2}{2} + o(x^2)\right) = -\frac{x^3}{2} + o(x^3)$$

Combining all three parts in the numerator:

$$\begin{aligned}\cosh(\alpha x) - e^{x^2} + x\log(\cos(x)) &= \left(1 + \frac{\alpha^2}{2}x^2 + o(x^3)\right) - \left(1 + x^2 + o(x^3)\right) - \frac{x^3}{2} + o(x^3) \\ &= \left(\frac{\alpha^2}{2} - 1\right)x^2 - \frac{1}{2}x^3 + o(x^3)\end{aligned}$$

Whole limit:

$$\lim_{x\to 0^+} \frac{\cosh(\alpha x) - e^{x^2} + x\log(\cos(x))}{x - \sin(x) + e^{-1/x^2}} = \lim_{x\to 0^+} \frac{\left(\frac{\alpha^2}{2} - 1\right)x^2 - \frac{1}{2}x^3 + o(x^3)}{\frac{1}{6}x^3 + o(x^3)}$$

Now we split by cases depending on whether the order-2 term in the numerator vanishes ($\frac{\alpha^2}{2} - 1 = 0 \iff \alpha = \pm\sqrt{2}$):

- **Case** $\alpha = \pm\sqrt{2}$ (the x^2 term cancels out, leaving order 3 on both top and bottom):

$$\begin{aligned}\lim_{x\to 0^+} \frac{-\frac{1}{2}x^3 + o(x^3)}{\frac{1}{6}x^3 + o(x^3)} &= \lim_{x\to 0^+} \frac{x^3\left(-\frac{1}{2} + \frac{o(x^3)}{x^3}\right)}{x^3\left(\frac{1}{6} + \frac{o(x^3)}{x^3}\right)} \\ &= \frac{-1/2}{1/6} = -3\end{aligned}$$

- **Case** $\alpha \neq \pm\sqrt{2}$ (the order-2 term dominates the numerator):
  Factor out x^2 from numerator and denominator:

$$\begin{aligned}\lim_{x\to 0^+} \frac{x^2\left[\left(\frac{\alpha^2}{2} - 1\right) - \frac{1}{2}x + \frac{o(x^3)}{x^2}\right]}{x^2\left[\frac{1}{6}x + \frac{o(x^3)}{x^2}\right]} &= \lim_{x\to 0^+} \frac{\left(\frac{\alpha^2}{2} - 1\right) - \frac{1}{2}x + o(x)}{\frac{1}{6}x + o(x)} \\ &= \frac{\frac{\alpha^2}{2} - 1}{0^+} = \begin{cases} +\infty & \text{if } \alpha < -\sqrt{2} \text{ or } \alpha > \sqrt{2} \quad \left(\frac{\alpha^2}{2} - 1 > 0\right) \\ -\infty & \text{if } -\sqrt{2} < \alpha < \sqrt{2} \quad \left(\frac{\alpha^2}{2} - 1 < 0\right) \end{cases}\end{aligned}$$

So the final result is:

$$\lim_{x\to 0^+} \frac{\cosh(\alpha x) - e^{x^2} + x\log(\cos(x))}{x - \sin(x) + e^{-1/x^2}} = \begin{cases} +\infty & \text{if } \alpha < -\sqrt{2} \text{ or } \alpha > \sqrt{2} \\ -3 & \text{if } \alpha = \pm\sqrt{2} \\ -\infty & \text{if } -\sqrt{2} < \alpha < \sqrt{2} \end{cases}$$

7) Lesson 13/11/26

7.1) Differentiability

Let's recap some theory

Definition (Differentiable on right/left)**.**

Let f be defined in a right/left neighborhood of $x_0 \in \mathbb{R}$. We say f is **differentiable on the right/left at** x_0 if the right/left limit of the difference quotient $\frac{\Delta f}{\Delta x}$ between x_0 and x exists finite for $x \to x_0$.
The real number

$$f'_{+/-}(x_0) = \lim_{x \to x_0^{+/-}} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta x \to 0^{+/-}} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

is the right/left derivative of f at x_0

Moreover we can say that a map is differentiable iff:

Theorem (Differentiability)**.**

A map f defined around a point $x_0 \in \mathbb{R}$ is differentiable at x_0 iff it is differentiable on both sides at x_0 and the left left and right derivatives coincide, in which case

$$f'(x_0) = f'_+(x_0) = f'_-(x_0)$$

Alternatively, if f is continuous at x_0 and differentiable at all points $x \neq x_0$ in a neighborhood of x_0. Then f is differentiable at x_0 provided that the limit of $f'(x)$ for $x \to x_0$ exists finite. If so

$$f'(x_0) = \lim_{x \to x_0} f'(x)$$

Where the continuity at x_0 must be verified

7.2) Exercises on Differentiability

b) Sin and abs value

Discuss differentiability at the point $x_0 = 0$:

$$f(x) = \sin|x|$$

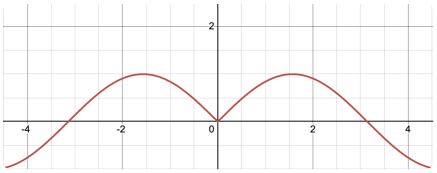
Using the definition we have

$$f'_+(0) = \lim_{x \to 0^+} \frac{\sin|x| - \sin(0)}{x - 0} = \lim_{x \to 0^+} \frac{\sin x}{x} = 1$$

However

$$f'_-(0) = \lim_{x \to 0^-} \frac{\sin|x| - \sin(0)}{x - 0} = \lim_{x \to 0^-} \frac{\sin -x}{x} = \lim_{x \to 0^-} \frac{-\sin x}{x} = -1$$

Since the limit is not the same, the function is not differentiable at $x_0 = 0$



Plot

c) Piecewise function

Discuss differentiability at the point $x_0 = 0$:

$$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Recall

$$\text{differentiable at } x_0 \implies \text{continuous at } x_0$$

which is equivalent to

$$\text{not continuous at } x_0 \implies \text{not differentiable at } x_0$$

A piecewise function is not necessarily continuous, let's check the continuity at x_0 first:

$$\lim_{x \to 0^+} e^{-1/x^2} = \lim_{x \to 0^-} e^{-1/x^2} = 0 = f(0)$$

Since it is continuous we must check the differentiability using the definition:

$$f'_+(0) = \lim_{x \to 0^+} \frac{e^{-1/x^2} - 0}{x - 0} = \lim_{x \to 0^+} \frac{e^{-1/x^2}}{x}$$

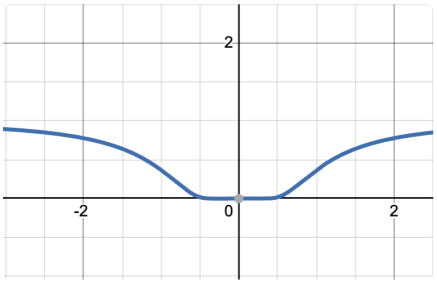
via substitution $t = \frac{1}{x}$ as $x \to 0^+$ and since exponential growth dominates linear growth

$$f'_+(0) = \lim_{t \to \infty} \frac{t}{e^{t^2}} = 0^+$$

Now the left limit:

$$f'_-(0) = \lim_{x \to 0^-} \frac{e^{-1/x^2} - 0}{x - 0} = \lim_{x \to 0^-} \frac{e^{-1/x^2}}{x} = \lim_{t \to -\infty} \frac{t}{e^{t^2}} = 0^-$$

Since both limits are equal the derivative at x_0 exists and is equal to 0.



Plot

d) Sqrt

Discuss differentiability at the point $x_0 = -1$:

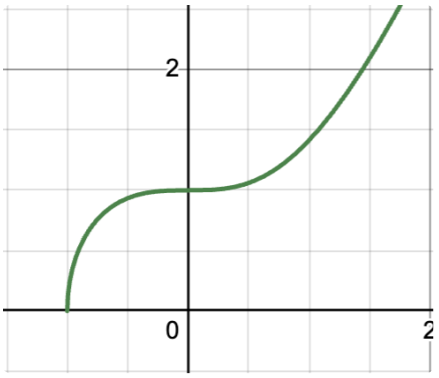
$$f(x) = \sqrt{1 + x^3}$$

Notice that the domain is $x \geq -1$ so we need to check only at -1^+

Using the definition we have

$$\begin{aligned}\lim_{x\to -1^+}\frac{\sqrt{1+x^3}-\sqrt{1-1}}{x-(-1)} &= \lim_{x\to -1^+}\frac{\sqrt{1+x^3}}{x+1}\\ &= \lim_{x\to -1^+}\frac{\sqrt{1+x^3}}{x+1}\frac{\sqrt{1+x^3}}{\sqrt{1+x^3}}\\ &= \lim_{x\to -1^+}\frac{1+x^3}{(x+1)\sqrt{1+x^3}}\\ &= \lim_{x\to -1^+}\frac{(x+1)(1-x+x^2)}{(x+1)\sqrt{1+x^3}}\\ &= \frac{3}{0^+} = +\infty\end{aligned}$$

Since the limit is not a finite real number, the function is not differentiable at $x_0 = -1$.



plot

7.3) Derivatives

Here is a list of the various derivatives of elementary functions

$f(x)$	$f'(x)$
x^α	$\alpha x^{\alpha-1}, \quad \forall \alpha \in \mathbb{R}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$1+\tan^2 x = \dfrac{1}{\cos^2 x}$
$\arcsin x$	$\dfrac{1}{\sqrt{1-x^2}}$
$\arccos x$	$-\dfrac{1}{\sqrt{1-x^2}}$
$\arctan x$	$\dfrac{1}{1+x^2}$
a^x	$(\log a)\, a^x$
$\log_a \lvert x\rvert$	$\dfrac{1}{(\log a)\, x}$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$

And some differentiation rules

$$\Big(\alpha f(x) + \beta g(x)\Big)' = \alpha f'(x) + \beta g'(x)$$
$$\Big(f(x)g(x)\Big)' = f'(x)g(x) + f(x)g'(x)$$
$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{\Big(g(x)\Big)^2}$$
$$\Big(g(f(x))\Big)' = g'(f(x))f'(x)$$

When using a substitution the last differentiation rule (chain rule) must be applied

7.4) Exercises on derivatives

c) Piecewise function

Say where the following maps are differentiable and find the derivatives

$$f(x) = \begin{cases} x^2 + 1 & x \geq 0 \\ e^x - x & x < 0 \end{cases}$$

First verify the continuity:
Both parts of the definition are continuous, so we must just look at 0

$$\lim_{x \to 0^+} x^2 + 1 = \lim_{x \to 0^-} e^x - x = 1 = f(0)$$

Therefore it is continuous

Now compute the derivative for both definitions:

$$x^2 + 1 \overset{x^\alpha = \alpha x^{\alpha-1}}{\longrightarrow} 2x$$
$$e^x - x \overset{a^x = \log(a)a^x}{\longrightarrow} \log(e)e^x - 1 = e^x - 1$$

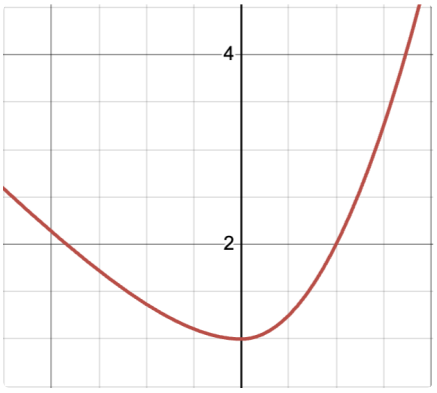
so we have

$$f'(x) = \begin{cases} 2x & x \geq 0 \\ e^x - 1 & x < 0 \end{cases}$$

Where the "$\geq$" is to be verified, as differentiability at 0 is not certain.
Both definitions are continuous, so differentiability needs only to be checked at $x_0 = 0$
Instead of using the definition, we use the theorem ($f'(x) = \lim_{x \to x_0} f'(x)$):

$$\lim_{x \to 0^+} 2x = \lim_{x \to 0^-} e^x - 1 = 0$$

So this function is fully differentiable.



Plot

a) Absolute value

Discuss where the following function is differentiable and find its derivative:

$$f(x) = x\sqrt{|x|}$$

The function is clearly continuous on all of $\mathbb{R}$, including at $x_0 = 0$.
One might suspect that since $|x|$ has a corner point at 0 (and $\sqrt{|x|}$ has a cusp), f might not be differentiable at $x_0 = 0$. Let's verify this.

Differentiate the function using the piecewise definition of absolute value:
First, rewrite $f(x)$ without the absolute value:

$$f(x) = x\sqrt{|x|} = \begin{cases} x\sqrt{x} & x \geq 0 \\ x\sqrt{-x} & x < 0 \end{cases} = \begin{cases} x^{3/2} & x \geq 0 \\ -(-x)^{3/2} & x < 0 \end{cases}$$

Now we differentiate both branches on the open intervals $x > 0$ and $x < 0$:

$$f'(x) = \begin{cases} \frac{3}{2}x^{1/2} & x > 0 \\ \frac{3}{2}(-x)^{1/2} & x < 0 \end{cases} = \begin{cases} \frac{3}{2}\sqrt{x} & x > 0 \\ \frac{3}{2}\sqrt{-x} & x < 0 \end{cases} = \frac{3}{2}\sqrt{|x|} \qquad \forall x \neq 0$$

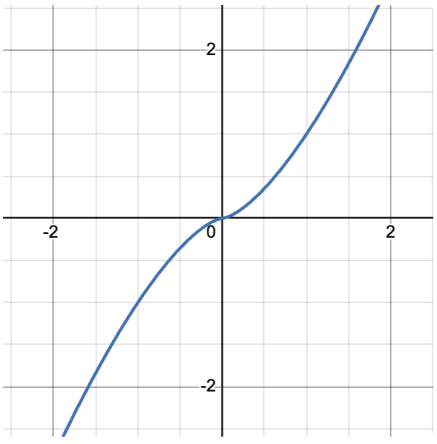
Here, the second branch ($x < 0$) was differentiated via the substitution $u(x) = -x$. Letting $\phi(u) = -u^{3/2}$, we are differentiating the composite function $\phi(u(x))$, which requires the chain rule:

$$\Big(\phi(u(x))\Big)' = \phi'(u(x)) \cdot u'(x) = -\frac{3}{2}u^{1/2} \cdot (-1) = \frac{3}{2}(-x)^{1/2}$$

Now we check differentiability at $x_0 = 0$. Since f is continuous at 0, we check the limits of $f'(x)$:

$$\lim_{x \to 0^+} \frac{3}{2}\sqrt{x} = \lim_{x \to 0^-} \frac{3}{2}\sqrt{-x} = 0$$

Thus f is differentiable at $x_0 = 0$ with $f'(0) = 0$! Our initial intuition was wrong (multiplying by x "smoothed out" the cusp at the origin), and $f'(x) = \frac{3}{2}\sqrt{|x|}$ holds for all $x \in \mathbb{R}$.



Plot

There is a second way (besides the definition) to calculate this limit:

First, as a preliminary exercise, consider the absolute value function:

$$a(x) = |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

This is not differentiable at 0 since:

$$a'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases} \quad \Longrightarrow \quad \lim_{x \to 0^+} a'(x) \neq \lim_{x \to 0^-} a'(x)$$

Notice that for $x \neq 0$, $a'(x)$ is precisely $\text{sign}(x) = \frac{|x|}{x}$.

Now we can compute $f'(x)$ for $x \neq 0$ using the product rule $\Big(h(x)g(x)\Big)' = h'(x)g(x) + h(x)g'(x)$ with $h(x) = x$ and $g(x) = \sqrt{|x|}$.
Since $g(x)$ is a composite function, we apply the chain rule $\Big(\psi(a(x))\Big)' = \psi'(a(x))a'(x)$ with $\psi(y) = \sqrt{y}$ and $a(x) = |x|$:

$$g'(x) = \frac{1}{2}|x|^{-1/2} \cdot \text{sign}(x) = \frac{1}{2\sqrt{|x|}} \frac{|x|}{x} = \frac{\sqrt{|x|}}{2x} \qquad \forall x \neq 0$$

Substituting this into the product rule for $x \neq 0$:

$$f'(x) = 1 \cdot \sqrt{|x|} + x \cdot \frac{\sqrt{|x|}}{2x} = \sqrt{|x|} + \frac{1}{2}\sqrt{|x|} = \frac{3}{2}\sqrt{|x|} \qquad \forall x \neq 0$$

Finally, since f is continuous at 0 and $\lim_{x \to 0} f'(x) = \lim_{x \to 0} \frac{3}{2}\sqrt{|x|} = 0$, we again conclude that $f'(0) = 0$, matching our first result.

Nested functions

Discuss where the following function is differentiable and find its derivative:

$$f(x) = \cos\left(e^{x^2+1}\right)$$

Since $f(x)$ is a double composition, we can break it down by introducing intermediate variables from the innermost function to the outermost:

- Innermost function: $u(x) = x^2 + 1 \overset{(x^\alpha)' = \alpha x^{\alpha-1}}{\longrightarrow} u'(x) = 2x$
- Middle function: $v(u) = e^u \overset{(a^u)' = (\log a)a^u}{\longrightarrow} v'(u) = (\log e)e^u = e^u$
- Outermost function: $g(v) = \cos v \overset{(\cos v)' = -\sin v}{\longrightarrow} g'(v) = -\sin v$

Now we apply the chain rule $\Big(g(v(x))\Big)' = g'(v(x)) \cdot v'(x)$ twice. First, for the outer layer:

$$f'(x) = g'\Big(v(u(x))\Big) \cdot \Big(v(u(x))\Big)' = -\sin\left(e^{x^2+1}\right) \cdot \left(e^{x^2+1}\right)'$$

Next, we apply the chain rule a second time to differentiate $v(u(x)) = e^{x^2+1}$:

$$\left(e^{x^2+1}\right)' = v'\Big(u(x)\Big) \cdot u'(x) = e^{x^2+1} \cdot (x^2+1)' = e^{x^2+1} \cdot 2x$$

Putting both pieces together, we obtain:

$$f'(x) = -\sin\left(e^{x^2+1}\right) \cdot e^{x^2+1} \cdot 2x = -2xe^{x^2+1}\sin\left(e^{x^2+1}\right) \qquad \forall x \in \mathbb{R}$$

Additionally one can compute it just in one run using the nested chain rule:
Once you are comfortable with the mechanism, you can apply the three-function chain rule $\Big(g(v(u(x)))\Big)' = g'(v(u(x))) \cdot v'(u(x)) \cdot u'(x)$ in a single line by differentiating from the outermost layer to the innermost:

$$\begin{aligned} f'(x) &= \left(\cos\left(e^{x^2+1}\right)\right)' \\ &= -\sin\left(e^{x^2+1}\right) \cdot \left(e^{x^2+1}\right)' \\ &= -\sin\left(e^{x^2+1}\right) \cdot e^{x^2+1} \cdot (x^2+1)' \\ &= -2xe^{x^2+1}\sin\left(e^{x^2+1}\right) \qquad \forall x \in \mathbb{R} \end{aligned}$$

General case (n nested functions)
By induction, let $f_1, f_2, \ldots, f_n$ be functions such that the composite function

$$F(x) = (f_n \circ f_{n-1} \circ \cdots \circ f_2 \circ f_1)(x) = f_n\Big(f_{n-1}\Big(\ldots f_2(f_1(x))\ldots\Big)\Big)$$

is well-defined. If f_1 is differentiable at x, f_2 is differentiable at $f_1(x)$, …, and f_n is differentiable at $f_{n-1}(\ldots f_1(x)\ldots)$, then F is differentiable at x and its derivative is the product of the derivatives of each layer, evaluated at everything inside it ("peeling the onion" from the outside in):

$$\begin{aligned} F'(x) &= f_n'\Big(f_{n-1}\Big(\ldots f_1(x)\ldots\Big)\Big) \cdot f_{n-1}'\Big(f_{n-2}\Big(\ldots f_1(x)\ldots\Big)\Big) \cdots f_2'\Big(f_1(x)\Big) \cdot f_1'(x) \\ &= f_1'(x) \cdot \prod_{k=2}^{n} f_k'\Big((f_{k-1} \circ \cdots \circ f_1)(x)\Big) \end{aligned}$$

8) Symbols

- $\implies$: implies, sufficient condition ($P \implies Q$ means P is sufficient for Q, Q is necessary for P).
- $\impliedby$: implied by, necessary condition
- $\iff$: necessary + sufficient, if and only if "iff"
- $\emptyset$: empty set
- $\mathbb{N}$: natural numbers; $\{0, 1, 2, ...\}$
- $\mathbb{Z}$: integer numbers; $\{..., -2, -1, 0, 1, 2, ...\}$
- $\mathbb{Q}$: rational numbers; $\left\{ \frac{m}{n} \; m, n \in \mathbb{Z}, n \neq 0 \right\}$
- $\mathbb{R}$: real numbers
- $\mathbb{C}$: complex numbers
- $\cup$: Union; or
- $\cap$: Intersection; and
- $\exists$: there exists

9) Taylor Expansions

Here is a list of some of the most frequent expansions

$$e^x = 1 + x + \frac{x^2}{2} + \cdots + \frac{x^k}{k!} + \cdots + \frac{x^n}{n!} + o(x^n)$$

$$\log(1+x) = x - \frac{x^2}{2} + \cdots + (-1)^{n-1}\frac{x^n}{n} + o(x^n)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^m \frac{x^{2m+1}}{(2m+1)!} + o(x^{2m+2})$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \cdots + (-1)^m \frac{x^{2m}}{(2m)!} + o(x^{2m+1})$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2m+1}}{(2m+1)!} + o(x^{2m+2})$$

$$\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{4!} + \cdots + \frac{x^{2m}}{(2m)!} + o(x^{2m+1})$$

$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \cdots + \left|\binom{-\frac{1}{2}}{m}\right| \frac{x^{2m+1}}{2m+1} + o(x^{2m+2})$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^m \frac{x^{2m+1}}{2m+1} + o(x^{2m+2})$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \cdots + \binom{\alpha}{n}x^n + o(x^n)$$

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-1)^n x^n + o(x^n)$$

$$\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + o(x^3)$$